

CRAFT: COMPOSITIONAL REWARD ALIGNMENT FOR TARGET-HIDDEN LOOP VERIFICATION

Anonymous authors

Paper under double-blind review

ABSTRACT

Valid loop invariants are closed under conjunction: useful clauses from different responses can prove a target even when no response does so alone. Yet standard training scores each response as an indivisible answer. We study this mismatch under *target-hidden loop verification*, where target assertions and postconditions Q are withheld during generation and training. CRAFT aligns both stages through rollout-combine-filter (RCF). At inference, RCF extracts and unions clauses, then uses syntax filtering and Houdini to retain an inductive subset before restoring Q . SFT distills a filtered multi-response union into one response. RL assigns credit within a rollout to its verifier-retained subset and across rollouts to complementary coverage beyond peers. Its target-independent strength signal comes from reachable executions and conservatively filtered synthetic negatives. On 832 C programs, five gpt-5-nano rollouts verify 74.76% of tasks, versus 61.78% for the strongest external baseline under each system’s fixed native budget. A paired Qwen3-8B before/after RL evaluation is in progress against the updated base checkpoint (pass@1 3.55%, combine@10 36.78%).

1 INTRODUCTION

Loop invariants turn unbounded executions into finite proofs, but a formally valid invariant can still be useless. An invariant is *valid* if it holds when execution first reaches the loop and remains true after every loop iteration. A verifier can check these obligations, yet its pass/fail outcome provides no graded measure of how much the invariant says: *true* is valid for every loop but proves nothing about its behavior or desired outcome. A useful invariant must also be strong enough to establish downstream properties. Since the exact reachable-state set may be expensive to compute or inexpressible in the annotation language, generation must navigate many valid but weak over-approximations and receive credit for approaching stronger ones.

This search need not succeed in a single attempt. Validity is closed under conjunction: the conjunction of two valid invariants is valid and no weaker than either one. Useful clauses can therefore be accumulated across samples instead of demanding a complete answer from one response. Model responses expose such partial results as clauses: one may discover a range bound, another a relational equality, and together they may prove a target that neither proves alone. Clause-wise verification can discard invalid clauses without losing the rest. The natural unit of inference is therefore the filtered clause pool, not a complete response.

Response-level rewards are mismatched to this unit in two ways. First, verifiable rewards credited to whole responses chiefly sharpen the sampling distribution toward responses the policy already produces, improving pass@1 without widening what k samples can cover (Yue et al., 2025). For a clause pool, such sharpening toward a few self-sufficient responses is at best orthogonal and at worst collapses the diversity that composition exploits. Second, a complete response is also the wrong unit of training credit. A whole-response reward withholds credit from useful clauses whenever another clause in the same response fails. A reward computed independently for each response cannot distinguish a constraint that expands group coverage from one already repeated by every peer. Under repeated sampling, this can encourage the policy to reproduce easy high-reward clauses instead of exploring a clause pool whose members work together. Our central claim is that loop-invariant learning should optimize the verified composition obtained from a fixed rollout group, rather than treating every rollout as an independent final answer.

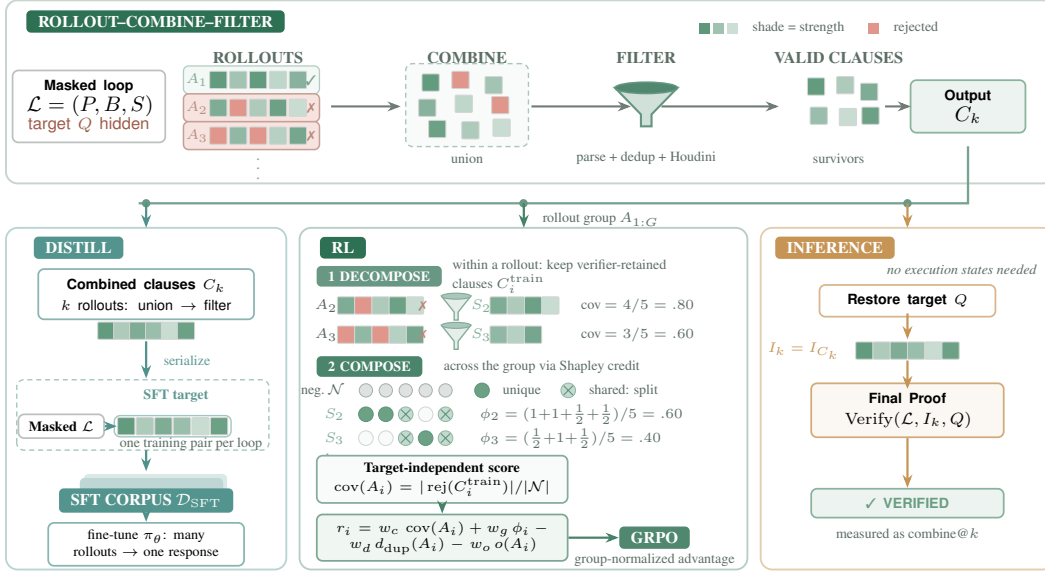


Figure 1: **Rollout-combine-filter is the organizing principle.** Target-hidden rollouts are parsed, unioned, and Houdini-filtered. SFT distills the combined set. RL *decomposes* each rollout onto its verifier-retained clauses—a dense, target-independent strength score from execution-derived negatives—and *composes* credit across the rollout group through a closed-form Shapley allocation, so a rejection already supplied by peers earns less. Executions construct and clean only training-time reward signals; inference restores Q only for the final proof.

We study this claim under *target-hidden loop verification*. The model sees the function preconditions, executable prefix, loop guard, and body, but the target set Q is withheld until the generated invariant set has been fixed. The final test still uses the original task: after Q is restored, the verifier must establish the loop obligations and prove every target at its original program point. Hiding Q prevents a model from copying a target that happens to be valid or using target-specific proof failures to steer generation. It also rules out target success as a training reward, making a graded, target-independent measure of invariant strength necessary.

CRAFT organizes inference around *rollout-combine-filter* (RCF). Given a budget of k target-hidden responses, RCF extracts each invariant annotation as a clause, unions clauses across responses, removes syntax failures, and applies Houdini to retain a jointly valid subset. Only then is Q restored for the final proof. We measure this inference-time object with $\text{combine}@k$: the fraction of programs verified after combining the first k responses. Unlike $\text{pass}@k$, which requires at least one response to contain the complete proof, $\text{combine}@k$ preserves useful partial answers. The objective is to make this curve both higher and faster to converge, so a small rollout budget recovers the combinations otherwise found only through much broader sampling.

Training aligns credit with the two levels of this inference procedure. *Within* each rollout, we project its extracted clauses onto a verifier-retained subset; one rejected clause does not erase credit earned by the others. We score the strength of that subset using the execution-derived negative examples it rejects, without adding a reward merely for well-formed syntax or whole-response validity. *Across* rollouts, we apply a closed-form Shapley allocation to the union of their independently retained negative-example coverage: repeated rejections share credit, whereas a rejection supplied by fewer peers receives more. Thus the reward of one rollout depends on what the other rollouts in its Group Relative Policy Optimization (GRPO) group discover (Shao et al., 2024). Synthetic supervised fine-tuning (SFT) additionally distills a filtered multi-response union into one individual response. Together, these stages make sampled responses more useful to the clause pool that RCF combines at inference.

Prior systems synthesize and prune candidate invariants at inference time, while learning-based methods train from verified outputs, solver feedback, or behavioral examples (Flanagan & Leino, 2001; Kamath et al., 2023; Cao et al., 2025; Si et al., 2020; Yan et al., 2025; Yang et al., 2026a;

Huang et al., 2026). We instead train for the filtered clause pool used at inference: credit preserves useful clauses within a response and depends on complementary coverage across responses.

Our experiments deliberately separate standalone success from combined inference. On 832 integer C programs, five target-hidden `gpt-5-nano` responses with union and Houdini verify 622 programs (74.76%), compared with 514 (61.78%) for the strongest external baseline under each system’s fixed native budget. The paired Qwen3-8B before/after RL evaluation is pending re-evaluation against the updated Base 8B checkpoint (pass@1 3.55%, combine@10 36.78%, $k_{95} = 30$); Section 5.5 reports the current baseline.

Contributions. This paper makes three contributions:

- We formulate target-hidden loop verification around a fixed-budget rollout–combine–filter objective. The evaluated object at inference is the verified composition measured by `combine@k`, not an isolated response.
- We align learning credit with that object in both directions: the reward preserves useful clauses within imperfect rollouts and allocates complementary coverage across rollouts, without using Q .
- We show that this alignment improves combined verification and rollout efficiency, and audit the execution-derived examples that provide its graded strength signal.

2 PRELIMINARIES

Loop verification. A task consists of a loop $\mathcal{L} = (P, B, S)$ and a target set $Q = \{(\ell, q_\ell)\}$, where predicate q_ℓ occurs at assertion or postcondition location ℓ . Here P describes the loop-entry states induced by the function preconditions and executable prefix, B is the guard, and S is the body. Let $\text{Reach}(\mathcal{L})$ be the reachable loop-head states, including the final guard test. An invariant I is *valid* (or inductive) when

$$P \Rightarrow I, \quad \{I \wedge B\} S \{I\},$$

which ensures $\text{Reach}(\mathcal{L}) \subseteq \llbracket I \rrbracket$ (Floyd, 1967; Hoare, 1969). The semantic ideal is an exact invariant I^* with $\llbracket I^* \rrbracket = \text{Reach}(\mathcal{L})$, although this set need not be finitely expressible in the annotation language.

For predicates I and J , write

$$I \preceq J \iff I \Rightarrow J \iff \llbracket I \rrbracket \subseteq \llbracket J \rrbracket.$$

Thus $I \preceq J$ means that I is at least as strong as J , and $I \prec J$ means that it is strictly stronger. For a finite clause set C , define

$$I_C \triangleq \bigwedge_{c \in C} c, \quad I_\emptyset \triangleq \text{true}.$$

We call C *jointly inductive* when I_C is valid, equivalently when

$$P \Rightarrow I_C, \quad \{I_C \wedge B\} S \{I_C\}.$$

We reserve $H_{\mathcal{L}}(C)$ for the ideal Houdini fixed point under exact establishment and preservation decisions, and $\text{Filter}_{\mathcal{L}}(C)$ for the implemented parsing, deduplication, and resource-bounded WP/Houdini pipeline.

A valid invariant is *Q-sufficient* when annotating the loop with I lets the verifier prove every q_ℓ at ℓ . For one post-loop target this reduces to $I \wedge \neg B \Rightarrow q$. Thus validity is target-independent, whereas sufficiency is tested only after restoring Q .

Target-hidden verification. The generator and all intermediate scores use $\mathcal{L} = (P, B, S)$, not Q . Assertions, postconditions, executable assertion calls, and conventional error-label names are masked; non-contract comments, including source-provenance paths, are removed. Preconditions and executable loop semantics remain visible. The generated invariant is fixed before the untouched program is restored for final verification. We study scalar-integer C programs with one loop; Appendix A gives the scope and a motivating nonlinear example.

3 RELATED WORK

Candidate generation, filtering, and prompt-time composition. Classical invariant synthesis uses abstract interpretation, affine relations, templates, interpolation, recurrences, syntax-guided search, or counterexamples; execution-driven systems instead mine likely properties from traces (Cousot & Cousot, 1977; Karr, 1976; Colón et al., 2003; McMillan, 2003; Kincaid et al., 2017; Alur et al., 2013; Garg et al., 2014; Ernst et al., 2007). Houdini iteratively removes invalid candidates to retain an inductive subset (Flanagan & Leino, 2001). LLM workflows reuse this pattern: LOOPY couples proposals with Houdini and repair, IRANK ranks candidates, and CLAUSE2INV combines retained clauses while iteratively accumulating counterexamples (Kamath et al., 2023; Chakraborty et al., 2023; Cao et al., 2025). LAM4INV, ACINV, AUTOSPEC, SPECGEN, SESPEC, LEMUR, and BALDUR similarly organize generation with static analysis, symbolic execution, verification, or repair (Wu et al., 2024a; Liu et al., 2024b; Wen et al., 2024; Ma et al., 2025; Yang et al., 2026b; Wu et al., 2024b; First et al., 2023). These are inference workflows; CRAFT fixes all target-hidden rollouts before joint filtering, receives no verifier feedback between calls, and trains responses for fixed-group composition.

Specialized neural invariant learning. CODE2INV learns a construction policy from solver feedback; CLN2INV and G-CLN learn continuous representations of linear and nonlinear relations; and LIPUS combines RL-based pruning with SMT solving (Si et al., 2018; 2020; Ryan et al., 2020; Yao et al., 2020; Yu et al., 2023). These task-specific models optimize one candidate or construction trajectory. We instead post-train a general language model so independently sampled responses contribute jointly under a fixed budget.

Behavioral-example objectives for specification strength. COINS assesses specifications using trusted positive and mutation-derived negative input–output cases (Yang et al., 2026a); SPECRL rewards complete method-level specifications that reject impossible input–output pairs (Huang et al., 2026); it neither targets loop invariants nor derives negatives from loop dynamics. Both show that behavioral examples distinguish weak but valid specifications from stronger ones. CRAFT supplies that loop-specific construction, filters each imperfect response to an inductive clause subset, and conditions its credit on negative states already rejected by peer rollouts.

Language-model training from formal feedback. INV BENCH and Wonda use verified data for invariant fine-tuning; CODERL uses execution signals; and RE:FORM applies verifier-guided RL to individual Dafny proofs (Wei et al., 2025; Pinto et al., 2026; Le et al., 2022; Yan et al., 2025). Unlike whole-output verifier rewards, CRAFT trains toward a filtered clause pool by salvaging the inductive part of an imperfect response and rewarding its strength and complementary group coverage. This distinction matters because verifiable-reward RL can improve small-budget accuracy without expanding broad-sampling coverage (Yue et al., 2025).

4 METHOD

CRAFT couples an inference protocol that proves targets by combining clauses across responses with a training pipeline that optimizes that protocol without ever seeing the target. Three design decisions organize this section: (i) *target hiding*—generation, distillation, and reward computation all operate on the masked loop, and the target is restored only for the final proof (Section 4.2); (ii) *the clause, not the response, is the credit unit* that survives filtering and earns reward (Section 4.1); (iii) *training mirrors inference*—credit rewards complementary coverage across a response group, because inference unions that group (Section 4.5). The guiding thesis is that because inference combines, training should optimize combinability: raising the combine@ k curve and reaching its large-pool performance with fewer rollouts.

4.1 ROLLOUT-COMBINE-FILTER

For a target-masked loop $\mathcal{L}$, the model emits ACSL loop-invariant annotations (Baudin et al., 2021). Parsing extracts each `loop invariant` ...; annotation as one clause; normalization turns response i into the clause sequence $A_i = \langle a_{i1}, \dots, a_{im_i} \rangle$. We cap it at K clauses ($K=20$; Appendix C), $\bar{A}_i = \langle a_{i1}, \dots, a_{i,\min(m_i, K)} \rangle$, with overflow $o(A_i) = \max(0, m_i - K)$. Extraction

Algorithm 1 Rollout–combine–filter

Input: masked loop $\mathcal{L}$, parsed clause sequences $A_{1:G}$, cap K
Output: filtered clause set C
1: $C \leftarrow \bigcup_i \text{set}(\text{Cap}_K(\text{Normalize}(A_i)))$
2: $C \leftarrow \text{ConservativeDedup}(C)$
3: **repeat**
4: $C_{\text{parse}} \leftarrow \text{ParserErrors}(\mathcal{L}, C); C \leftarrow C \setminus C_{\text{parse}}$
5: **until** $C_{\text{parse}} = \emptyset$ or $C = \emptyset$
6: **repeat**
7: $C_{\text{drop}} \leftarrow \{c \in C : \neg \text{WP}_{\text{est,pres}}(\mathcal{L}, C, c)\}; C \leftarrow C \setminus C_{\text{drop}}$
8: **until** $C_{\text{drop}} = \emptyset$ or $C = \emptyset$
9: **return** C

makes the annotation, rather than the complete response, the unit that can survive filtering and earn credit. We write $\text{set}(\bar{A}_i)$ for the underlying clause set when applying set operations.

Given G responses, RCF unions their clauses,

$$C_G^{\text{pool}} = \bigcup_{i=1}^G \text{set}(\bar{A}_i),$$

then iteratively removes syntax failures and applies Houdini to delete clauses whose establishment or preservation obligations fail (Flanagan & Leino, 2001). We write the resulting inductive subset as $\text{Filter}_{\mathcal{L}}(C_G^{\text{pool}})$. Conjoining valid clauses preserves validity and can combine a bound from one response with a relation from another; filtering salvages such clauses from otherwise imperfect responses but cannot invent a missing relation. Algorithm 1 computes $\text{Filter}_{\mathcal{L}}$; we write its fixed point as $H_{\mathcal{L}}(C)$ and the conjunction of a clause set C as $I_C = \bigwedge_{c \in C} c$. The algorithm is shared by SFT synthesis and inference.

The following statements formalize the composition used by RCF; all proofs are deferred to Appendix B.

Proposition 1 (Closure and strengthening under conjunction). *Let $I_1, \dots, I_m$ be valid invariants of the same loop, for finite $m \geq 1$. Then $I = \bigwedge_{j=1}^m I_j$ is valid and implies every I_j , i.e., $I \preceq I_j$. Moreover, $I \prec I_j$ exactly when $I_j \not\Rightarrow \bigwedge_{\ell \neq j} I_\ell$.*

Proposition 2 (Mutually supported composition). *Let $C = \{c_1, \dots, c_m\}$ be any finite set of candidate clauses. The conjunction I_C is a valid invariant whenever*

$$P \Rightarrow I_C, \quad \forall j, \{I_C \wedge B\} S \{c_j\}.$$

Individual obligations $\{c_j \wedge B\} S \{c_j\}$ need not hold: preservation may be supplied by any number of companion clauses in C . Mutual support cannot, however, repair a clause that fails establishment from P . The resulting valid invariant satisfies $I_C \preceq c_j$, with $I_C \prec c_j$ whenever $c_j \not\Rightarrow I_{C \setminus \{c_j\}}$.

Theorem 3 (Candidate-relative optimality of ideal Houdini). *For a finite parser-admitted pool C , assume exact decisions for all establishment and preservation obligations in Algorithm 1. Its fixed point $H_{\mathcal{L}}(C)$ is jointly inductive, and every jointly inductive subset $C' \subseteq C$ satisfies $C' \subseteq H_{\mathcal{L}}(C)$. Consequently, $I_{H_{\mathcal{L}}(C)} \preceq I_{C'}$: the result is the strongest conjunction expressible as a subset of C . This maximality is relative to the supplied pool and the exact-oracle assumption.*

This is the classic Houdini greatest-fixpoint property (Flanagan & Leino, 2001), restated in our clause notation. In practice the per-obligation prover budget (Section 5.1) makes this oracle approximate.

RL scoring additionally removes clauses contradicted by observed reachable states $\mathcal{P}$ before Houdini:

$$\text{PF}_{\mathcal{P}}(C) = \{c \in C : \forall p \in \mathcal{P}, p \models c\}.$$

This positive filter is a cheap refutation step, not a validity proof; Frama-C/WP remains the authority. It is used only for RL scoring; inference performs the same identifier-scope checks but no execution-based rejection. Appendix C gives normalization and filtering details.

4.2 INFERENCE OBJECTIVE

At inference, RCF samples k masked responses and produces

$$C_k^{\text{pool}} = \bigcup_{i=1}^k \text{set}(\bar{A}_i), \quad C_k = \text{Filter}_{\mathcal{L}}(C_k^{\text{pool}}), \quad I_k = I_{C_k}.$$

Only then is Q restored. The target utility and inference objective are

$$U_Q(A_{1:k}) = 1[\text{Verify}(\mathcal{L}, I_k, Q)], \quad \max_{\pi} \mathbb{E}_{A_{1:k} \sim \pi(\cdot|\mathcal{L})}[U_Q(A_{1:k})].$$

Here Verify requires establishment, preservation, and every restored target at its original location; for one post-loop target, its last obligation is $I_k \wedge \neg B \Rightarrow q$. Its empirical counterpart is `combine@k`, which can succeed even when no standalone response is sufficient. The training sections below therefore optimize a target-hidden surrogate for this curve.

4.3 TARGET-INDEPENDENT STRENGTH EXAMPLES

To grade inductive sets without Q , we execute a deterministic, precondition-checked input sweep and record reachable loop-head states $\mathcal{P}$. From these executions we construct three negative-example families: local one- or two-variable perturbations that break relations, continuations of the real body after a witnessed loop exit, and perturbations outside observed variable ranges. Cleaning is deliberately conservative—any candidate that executions cannot rule out is dropped: it removes observed reachable states, possible fresh entries, duplicates, and candidates involving unsupported state. These examples are training signals, not proofs of global unreachability; unsupported or incomplete executions disable the affected family. Figure 5, Algorithm 2, and exact budgets appear in Appendix C.1.

Let $\mathcal{N}$ be the retained negative traces. For a retained clause set C , $\text{rej}(C) \subseteq \mathcal{N}$ contains each trace on which at least one clause in C is false at some recorded state; singleton traces recover ordinary state rejection.

4.4 DISTILLING MULTI-RESPONSE SEARCH

For each masked loop $\mathcal{L}$ in the training corpus $\mathcal{D}_{\text{train}}$, let $C_{\mathcal{L}}^{\text{pool}}$ be a sampled response-group union and $C_{\mathcal{L}}^{\text{sft}} = \text{Filter}_{\mathcal{L}}(C_{\mathcal{L}}^{\text{pool}})$. We serialize every nonempty result as one SFT target:

$$\mathcal{D}_{\text{SFT}} = \{(\mathcal{L}, \text{serialize}(C_{\mathcal{L}}^{\text{sft}})) : \mathcal{L} \in \mathcal{D}_{\text{train}}, C_{\mathcal{L}}^{\text{sft}} \neq \emptyset\}.$$

Neither input nor target contains Q : SFT serializes a Houdini-retained multi-response search result as one response. The execution states of Section 4.3 are used only for RL rewards, never as SFT answers.

4.5 CLAUSE-DECOMPOSED, GROUP-COMPOSITIONAL REWARD

The reward design responds directly to the sharpening tendency described in Section 1. Group-relative optimization with a whole-response reward concentrates probability mass on a few high-reward responses, eroding the clause diversity that RCF combines at inference. We therefore decompose credit *within* each rollout—surviving clauses earn reward even when siblings fail—and re-allocate credit *across* the group by each rollout’s marginal contribution to pooled coverage, paying the policy for expanding the joint clause pool rather than for reproducing its most common clauses.

For rollout i , define its verifier-retained subset and rejected negatives as

$$C_i^{\text{pf}} = \text{PF}_{\mathcal{P}}(\text{set}(\bar{A}_i)), \quad C_i^{\text{train}} = \text{Filter}_{\mathcal{L}}(C_i^{\text{pf}}), \quad R_i = \text{rej}(C_i^{\text{train}}).$$

This projection implements *intra-rollout decomposition*: malformed or non-inductive clauses are removed without erasing useful clauses in the same response. When retained negatives exist, the reward adds no bonus merely for valid syntax or whole-response validity. The subset receives the dense, target-independent strength score

$$\text{cov}(A_i) = \frac{|R_i|}{|\mathcal{N}|}.$$

This and the following allocation apply when $\mathcal{N} \neq \emptyset$; when the sampler retains none, we use the binary fallback in Appendix C.2.

Independent coverage, however, rewards a complementary discovery exactly like one repeated by every peer. We therefore allocate credit for the union $\bigcup_i R_i$ across the rollout group $\mathcal{G} = \{1, \dots, G\}$ as a coverage game. Let $f(n) = \sum_{j=1}^G \mathbf{1}[n \in R_j]$ count how many peers reject trace n . The Shapley value of rollout i has the closed form (Shapley, 1953) (proof in Appendix B)

$$\phi_i = \frac{1}{|\mathcal{N}|} \sum_{n \in R_i} \frac{1}{f(n)}, \quad \sum_{i=1}^G \phi_i = \frac{|\bigcup_i R_i|}{|\mathcal{N}|}. \quad (1)$$

Thus a unique rejection receives full credit, whereas $f(n)$ redundant rejections split it. The Shapley allocation is the unique credit rule satisfying efficiency, symmetry, linearity, and the null-player axiom (Shapley, 1953); unlike a leave-one-out marginal, which credits only rejections supplied by no peer and does not account for the full union, it splits credit for shared rejections instead of discarding it. This closed form uses already-computed rejection sets and adds no verifier calls. It implements *inter-rollout composition*: the same response earns less group credit when its verified constraints are already supplied by peers.

With the within-response duplicate count $d_{\text{dup}}(A_i) = |\bar{A}_i| - |\text{dedup}(\bar{A}_i)|$ (clauses identical after normalization), the reward is

$$r_i = w_c \text{cov}(A_i) + w_g \phi_i - w_d d_{\text{dup}}(A_i) - w_o o(A_i), \quad (2)$$

where $(w_c, w_g, w_d, w_o) = (1.0, 0.3, 0.02, 0.05)$; Section 5.6 ablates each term. GRPO normalizes these group-conditioned rewards within the sampled response group. Shapley credit distinguishes equal-coverage rollouts when their rejection sets overlap differently, favoring less redundant coverage. It remains a target-independent behavioral surrogate rather than credit for final Q -sufficiency. Each training step samples a group of G responses per masked loop, scores them by Equation (2), and applies one GRPO update; this training-time group is distinct from the inference sweep over k (Section 4.2). Appendix C.2 gives the standard policy objective, duplicate normalization, and fallback behavior.

5 EXPERIMENTS

We ask how combination scales with sampling (RQ1), transfers across models (RQ2), compares with specialized tools (RQ3), changes after RL (RQ4), and depends on reward and sampler design (RQ5).

5.1 EXPERIMENTAL SETUP

Data and judge. The cleaned training artifacts contain 37,481 target-masked scalar-integer single-loop RL records; 3,200 form SFT targets. Evaluation uses the standard benchmark suites of prior learning-based invariant-generation systems: 832 C programs, comprising 316 linear programs (133 CODE2INV (Si et al., 2020), 84 SyGuS 2019 (Alur et al., 2013), 99 SV-COMP 2024 (Beyer, 2023)) and 50 nonlinear-arithmetic (NLA) programs (30 LIPuS (Yu et al., 2023), 20 SV-COMP 2024) as released with CLAUSE2INV (Cao et al., 2025), plus 466 integer programs from Loopy (Kamath et al., 2023). The overlap and prompt-sanitation audits are in Appendix F. We remove non-contract comments and mask all target predicates before generation, then restore the untouched source only after invariants are fixed. Success requires Frama-C/WP (Kirchner et al., 2015), under a 5-second per-obligation prover budget, to prove establishment, preservation, and every restored target; all failures count as zero.

Inference protocol. All CRAFT evaluation runs share target masking, the canonical prompt, clause processing, and Houdini; external tools retain their own target-hidden orchestration. Within CRAFT, API models use `reasoning_effort=none`; all locally served checkpoints use non-thinking decoding to match data synthesis and training. Decoding uses temperature 1.0 and top- p 1.0. API calls cap completions at 8,192 tokens, including provider-internal reasoning tokens; local vLLM calls cap each response at 2,048 emitted tokens. We use no re-roll or target feedback. RQ1 reuses one saved batch of 100 stochastic responses per task. Appendix G gives benchmark provenance, serving, and training details.

Metrics. Throughout evaluation, success means Q -sufficiency under the final verifier. $\text{pass}@k$ is the probability that at least one of k responses proves Q . Whenever a saved pool contains c successful responses among $n = 100$, we use the unbiased estimator $\widehat{\text{pass}@k} = 1 - \binom{n-c}{k} / \binom{n}{k}$ for that program and average it over programs. Only responses that prove Q count as successes. $\text{combine}@k$ is the fraction of programs proved by unioning the first k responses and applying Houdini. $\text{pass}@k$ requires one response to contain the whole proof, whereas $\text{combine}@k$ can combine clauses from different responses. We do not compute the exact all-subset analogue for $\text{combine}@k$, because it would require filtering $\binom{100}{k}$ unions per program; we evaluate the fixed saved prefix instead. For the measured budget grid $\mathcal{K} = \{1, 10, 30, 50, 100\}$, we report $k_{95} = \min\{k \in \mathcal{K} : \text{Comb}_\pi(k) \geq 0.95 \text{ Comb}_\pi(100)\}$, where $\text{Comb}_\pi(k)$ is the measured $\text{combine}@k$ curve. A smaller k_{95} means the retained pool reaches its large-budget performance sooner.

5.2 RQ1: HOW DO PASS@K AND COMBINE@K SCALE?

We sample 100 target-hidden responses per task from six base checkpoints: Qwen3-0.6B, Qwen3-4B, Qwen3-8B, Qwen3-14B, Qwen3-30B-A3B (Yang et al., 2025), and Llama 3.1 8B. Complete values and k_{95} appear in Appendix 8.

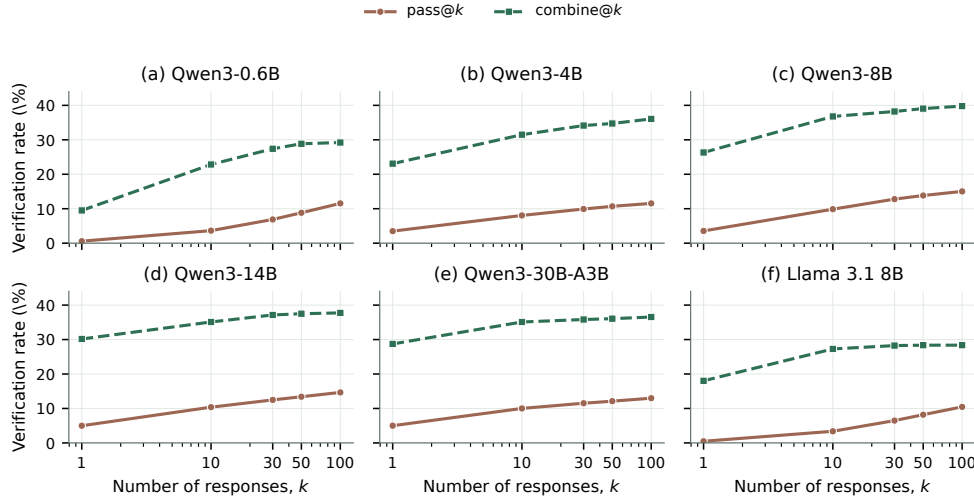


Figure 2: $\text{pass}@k$ (unbiased estimate) and prefix $\text{combine}@k$. Combination saturates by $k = 30$, while $\text{pass}@k$ keeps growing.

At $k = 1$, prefix combine is 8.92–25.18 points above the unbiased pass estimate across checkpoints. Because these are different estimands rather than paired first-response outcomes, we treat this gap as descriptive; RQ2 reports the paired effect of processing the same response. In the retained response pools, every model reaches 95% of $\text{combine}@100$ by $k = 50$, with at most 1.56 additional points through $k = 100$. In contrast, $\text{pass}@30$ reaches only 59.6–88.8% of $\text{pass}@100$ on the five Qwen checkpoints and gains another 1.46–4.66 points by $k = 100$. Llama 3.1 8B is harder in standalone proof generation— $\text{pass}@100$ is 10.46%—but shows the same compositional pattern: combine rises from 18.03% at $k = 1$ to 28.25% at $k = 30$, then changes by only 0.12 points. Combination thus front-loads useful coverage, whereas standalone proofs require wider sampling. The $\text{combine}@100$ ceilings (28.37–39.78%) further suggest that these pools remain bottlenecked by clause quality or diversity: extending the budget to 100 responses does not remove the gap.

Insight. $\text{pass}@1$ is not an appropriate primary metric for compositional invariant generation. By treating a response as indivisible, it assigns zero both to useless outputs and to incomplete outputs containing valid, reusable clauses. It remains a standalone diagnostic, whereas $\text{combine}@k$ directly measures whether clauses found within the inference budget form a sufficient invariant after filtering.

5.3 RQ2: HOW STRONG IS COMBINE@1 ACROSS MODELS?

The main neural experiment evaluates exactly one response per program. $\text{pass}@1$ treats that response as one all-or-nothing invariant set. $\text{combine}@1$ extracts invariant annotations as clauses, removes duplicates, and runs Houdini on the resulting pool. No $\text{pass}@k > 1$ or $\text{combine}@k > 1$ result is included in this RQ, so model quality is not confounded with a larger inference budget.

k	Verified (%)								Cost per task	
	Linear		NLA		Loopy		All		Time (s)	Tokens
	pass	comb	pass	comb	pass	comb	pass	comb		
<i>GPT-5-nano</i>										
1	2.53	49.68	0.00	8.00	4.72	45.71	3.61	44.95	29.7	1,136
10	17.72	67.72	4.00	10.00	19.74	71.24	18.03	66.23	190.4	4,322
<i>GPT-5</i>										
1	24.05	62.03	2.00	10.00	24.46	63.52	22.96	59.74	18.2	1,231
10	53.16	71.20	10.00	8.00	57.73	68.88	53.13	66.11	75.5	12,251
<i>Claude Sonnet-4.6</i>										
1	32.28	58.86	8.00	8.00	39.27	63.30	34.74	58.29	6.7	1,307
10	43.67	66.77	10.00	12.00	51.07	64.81	45.79	62.38	66.9	13,072
<i>DeepSeek V4-Flash</i>										
1	16.14	62.66	6.00	12.00	21.67	61.59	18.63	59.01	35.9	1,209
10	47.47	72.15	8.00	8.00	55.15	72.53	49.40	68.51	105.2	13,580

Table 1: Cross-model results by benchmark stratum (316 linear, 50 NLA, 466 Loopy, 832 overall programs). Each stratum reports $\text{pass}@k$ (pass) and $\text{combine}@k$ (comb) percentages side by side at the rollout budget k in the first column; within each combine column and budget block, the best value is in bold. Time covers the end-to-end $\text{combine}@k$ pipeline; tokens cover the k shared responses. reasoning_effort=none throughout; the $k=10$ rows use ten independent single-response calls per task. GPT-5-mini is omitted: it exposes no true reasoning-off setting, and even its lowest tier draws a stochastic, sometimes total, hidden-reasoning token budget on this benchmark, so no run of it satisfies the reasoning_effort=none protocol used for every other row (Appendix H.5 quantifies this and the separate thinking-mode leak we found and fixed for DeepSeek V4-Flash).

Because $\text{pass}@1$ and $\text{combine}@1$ reuse the same response, their paired change isolates the net effect of clause extraction, normalization, and Houdini without additional model calls. All rows use provider-reported token counts and full-workload means over the same 832 programs. Appendix 16 reports the paired rates by benchmark stratum.

5.4 RQ3: HOW DOES CRAFT COMPARE WITH SPECIALIZED TOOLS?

We compare against the target-hidden pipelines of AUTOSPEC, SESPEC, and CLAUSE2INV, and against Daikon 5.8.24 (Wen et al., 2024; Yang et al., 2026b; Cao et al., 2025; Ernst et al., 2007). All LLM-based rows use gpt-5-nano. Naive makes one call without filtering or refinement. Because $\text{pass}@1$ judges all clauses in a response as an indivisible conjunction, one invalid or non-inductive clause makes the whole response fail; CRAFT’s richer raw outputs can therefore score below the shorter Naive outputs before Houdini filtering. CLAUSE2INV’s unsupported inputs remain failures in the common 832-program denominator. Budgets are fixed before comparison but not matched: each system retains its predeclared native orchestration.

System	Linear / 316	NLA / 50	Loopy / 466	All / 832	Tokens / task	Time / task
AUTOSPEC	47.78%	6.00%	42.27%	42.19%	2,181.72	27.34 s
SESPEC	28.80%	4.00%	33.05%	29.69%	22,081.61	68.50 s
CLAUSE2INV	20.89%	0.00%	0.00%	7.93%	1,056.25	8.41 s
DAIKON	23.10%	0.00%	21.67%	20.91%	0	4.89 s
LOOPY	20.25%	0.00%	19.74%	18.75%	14,513.37	147.16 s
Naive	15.19%	2.00%	18.88%	16.47%	225.95	2.98 s
CRAFT (pass@1)	2.53%	0.00%	4.72%	3.61%	1,135.83	7.86 s
CRAFT (combine@1)	49.68%	8.00%	45.71%	44.95%	1,135.83	29.66 s
CRAFT (combine@5)	64.87%	12.00%	63.73%	61.06%	2,161.52	91.78 s
CRAFT (combine@10)	67.72%	10.00%	71.24%	66.23%	4,322.38	190.35 s

Table 2: Common target-hidden comparison with `reasoning_effort=none`, using the same untouched inputs and Frama-C/WP final judge. Tokens are mean totals per task.

The main table reports only the completed non-reasoning rerun. The archived reasoning-enabled comparison is retained in Appendix 17; no archived value is reused as a non-reasoning result. Under this setting, increasing CRAFT from five to ten candidates raises verification from 508/832 (61.06%) to 551/832 (66.23%).

Failure modes under target hiding. The specialized systems degrade for different structural reasons. In the evaluated one-round configuration, AUTOSPEC requests eight completions in parallel, pools their clauses, and verifier-filters the union; it is therefore already an eight-proposal combination method rather than a single-proposal baseline. Nevertheless, it verifies 351 programs, 23 fewer than CRAFT combine@1, while using 2,181.72 versus 1,135.83 tokens per task. Its extra samples mainly add redundant or weak clauses rather than the complementary facts recovered by CRAFT’s clause processing. SESPEC instead makes 5.09 sequential calls per completed task on average. This refinement-heavy schedule spends 22,081.61 tokens but does not provide the independent parallel exploration needed when the target is absent; moreover, errors from the target-hidden verifier diagnose syntax and inductiveness, not which missing relation would establish the hidden goal. CLAUSE2INV’s ICE loop normally uses false-state (`F :`) counterexamples induced by the postcondition. Target-hidden execution suppresses the postcondition check and retains only pre-state and transition counterexamples, removing its principal goal-directed feedback signal; its required transition VCs are also unavailable for all 466 Loopy programs.

LOOPY exhibits a separate interface failure. Its extractor accepts only the first valid Markdown-fenced code block. Although 98.8% of the non-reasoning initial responses contain textual `loop` invariant clauses, only 1.75% use the required fence and reach Houdini; for repair responses the corresponding rates are 94.5% and 1.13%. Houdini can prune bad clauses but cannot recover discarded ones. Target hiding further removes the assertion-specific checker diagnostic expected by Loopy’s repair prompt, leaving only generic inductiveness feedback: repair adds just 15 successes to the 141 initially solved tasks, for 156/832 (18.75%) overall. Thus this row primarily measures a brittle parser and a repair protocol mismatched to the hidden-target setting, rather than the quality of the invariants present in the raw responses. The archived medium-reasoning run is reported separately in the appendix.

Finding. Target visibility is a problem boundary, not an implementation detail. Goal-directed feedback—CLAUSE2INV’s postcondition-induced counterexamples, LOOPY’s assertion-specific diagnostics—vanishes once the target is hidden, and the two systems fall to 7.93% and 18.75%; five target-hidden responses combined by CRAFT verify 61.06%. Under target hiding, complementary combination survives where iterative goal-directed repair does not.

5.5 RQ4: HOW MUCH DOES REINFORCEMENT LEARNING IMPROVE INFERENCE?

We evaluate matched Qwen3-8B before–after RL paths from base and SFT initializations. Inference is fixed within each pair; RQ1 is not used to estimate the RL effect.

rl_panels.pdf placeholder — generated by figures/plot_rl_panels.py from the canonical RL pool artifact (three-seed means; pass@ k left, combine@ k right with the SFT combine@100 reference line and the crossing budget k^* annotated).

Figure 3: Response-level vs. compositional effects of RL (Qwen3-8B, SFT initialization). (a) pass@ k is expected to remain flat, consistent with redistribution rather than support expansion (Yue et al., 2025). (b) combine@ k shifts upward at small budgets: the RL policy reaches its initialization’s combine@100 (dotted line) at k^* samples, and combine@100 itself does not degrade.

What to expect. If RL with whole-response credit merely sharpens the distribution (Yue et al., 2025), it should raise pass@1 while leaving or degrading large- k coverage. Clause-decomposed, group-compositional credit predicts the opposite pattern: pass@ k stays flat, while combine@ k shifts up at small budgets without degrading at large ones.

The SFT stage alone moves the operating point substantially. Distilling the filtered multi-response union into single responses lifts Qwen3-8B from pass@1 3.55%/combine@1 26.32% to 25.98%/49.88%: one distilled response verifies more programs than one hundred base responses combined (combine@100 39.78%). The distilled 8B also matches or exceeds the distillation teacher itself: at $k=10$ it reaches pass 54.58%/combine 74.04%, against 45.79%/62.38% for Claude Sonnet-4.6 under the same protocol (Table 1). Full SFT curves across four backbones and per-GPU serving costs appear in Appendix H.2.

Initialization	Training stage	pass@1	combine@1	combine@10	combine@100	k_{95}
Base 8B	before RL (Zero)	3.55%	26.32%	36.78%	39.78%	30
Base 8B	after RL (RL-Zero)	TBD	TBD	TBD	TBD	TBD
SFT 8B	before RL (SFT)	25.98%	49.88%	74.04%	76.80%	10
SFT 8B	after RL (SFT+RL)	TBD	TBD	TBD	TBD	TBD

Table 3: Matched Qwen3-8B evaluation before and after RL.

The RL-Zero row awaits re-evaluation against the updated Base 8B baseline (pass@1 3.55%, combine@1 26.32%, combine@10 36.78%, $k_{95} = 30$).

Results. Consistent with the redistribution view, response-level accuracy does not improve after RL (Figure 3, left; pass@1 TBD vs. TBD). The redistribution is instead steered toward the complementary tail of the clause pool: combine@ k^* with $k^*=TBD$ matches the SFT checkpoint’s combine@100 (76.80%), an effective TBD-fold budget reduction, and combine@100 itself does not degrade (TBD vs. 76.80%). RL thus acts as a *diversity compressor*, packing into k^* samples the useful clauses its initialization surfaces only across one hundred. A pool-membership analysis against TBD initialization samples attributes part of the gain to clauses the initialization surfaces with probability below TBD per response. All TBD entries in this paragraph await the cleaned-data retraining and the canonical RL pool artifact.

5.6 RQ5: WHAT DRIVES THE TRAINING GAIN?

Both ablations use Qwen3-8B and three matched seeds with all other training and inference settings fixed; Appendix H.4 reports per-seed combine@10.

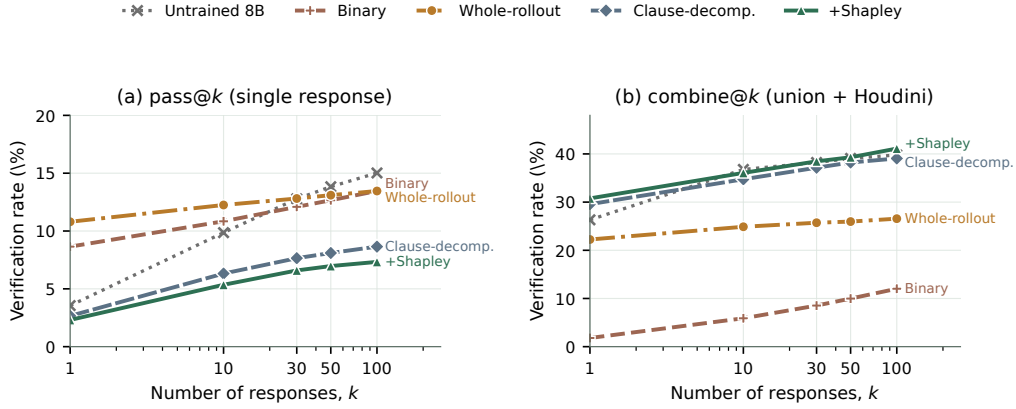


Figure 4: Reward-function ablation on Qwen3-8B (three-seed means). Binary and whole-rollout credit keep single-response accuracy high (a) but train redundant outputs, so their combine curves flatten (b); clause-decomposed and complementarity-aware credit shifts probability mass toward combinable clauses, lifting combine@ k above the untrained model at every k . The regularized variant (ours) awaits re-evaluation under the final training configuration; exact numbers for all variants appear in Table 13 (Appendix H.4).

Reward-function ablation. Variants progress from binary inductiveness to whole-response coverage, then add clause salvage, Shapley credit, and duplicate regularization. All use a 20-clause cap; coverage variants share the overflow cost, while the binary control remains $\{0, 1\}$.

Figure 4 reveals two effects (exact numbers in Table 13, Appendix H.4). *First, binary inductiveness collapses combination.* Although its pass@1 reaches 8.65%, the policy degenerates into repeating a few safe clauses, so combine@100 stalls at 12.02%, 27.8 points below the untrained model. This is the sharpening failure mode anticipated in Section 5.5: whole-response credit preserves standalone accuracy but collapses the pool; the remaining variants show that clause decomposition and group-compositional allocation are each needed to avoid it. *Second, credit must be decomposed to clauses.* Moving from whole-rollout to clause-decomposed coverage raises combine@100 from 26.56% to 39.06%, and Shapley credit adds a further +2.05 points (41.11%); rewarding complementary coverage across the group translates directly into combinable invariants, at the price of fragmenting individual responses (pass@1 drops to 2.31%). The regularized variant (ours) awaits re-evaluation under the final training configuration.

Finding. The reward shapes what can be *combined*, not only what is *proven*. Binary inductiveness trains repetition and whole-response credit trains redundant self-sufficient responses; clause-decomposed, complementarity-aware (Shapley) credit is what improves the union, lifting combine@100 from 12.02% to 41.11% across the ablation ladder.

Negative-sampling ablation. *Random* uses uniform local perturbations; *Structured* uses relational, post-exit, and range families. Both share executions, filters, seed, and a 512-trace maximum; cleaning may leave different realized sizes.

Negative sampler	Reward var.	Collision	pass@1	combine@1	combine@10	combine@100	k_{95}
Random	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Structured (ours)	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 4: Negative-sampler ablation under a shared maximum budget.

6 CONCLUSION AND LIMITATIONS

CRAFT makes filtered clause composition the inference target: RCF combines rollouts, SFT distills their union, and target-independent RL rewards useful clauses and complementary coverage.

The recipe transfers beyond loop invariants. It applies to problems whose answers (i) decompose into independently checkable units combined by conjunction, (ii) admit a cheap mechanical filter for unit validity, (iii) include valid-but-weak degenerate solutions that pass/fail cannot grade, and (iv) allow behavioral negatives, on which weak and strong solutions disagree, to be derived from executions or simulations. Test generation judged by mutant killing, auxiliary-lemma synthesis for proof assistants, and data-constraint discovery all satisfy these conditions.

Evidence is limited to scalar-integer, single-loop C programs, Frama-C/WP and ACSL, and vLLM-served local checkpoints. Credit also filters rollouts separately whereas inference filters their union. Appendices I and K detail these scope and reward approximations.

AI USE STATEMENT

Generative AI tools are integral to the evaluated system: they produce the invariant candidates and synthetic SFT data described in the paper. They were also used during research to refine hypotheses, methodology, and experiments; formulate and check mathematical claims; implement and test software; analyze and interpret experimental artifacts; prepare figures and tables; search and summarize literature; and draft, edit, and translate manuscript text. The authors reviewed all AI-assisted work and tested generated code. They take responsibility for independently validating the final content, empirical claims, and released artifacts.

REPRODUCIBILITY STATEMENT

The method and objective are specified in Section 4; Appendices C–G give the implementation defaults, prompt, training interface, data-overlap audit, and evaluation protocol. Appendix H organizes the available measurements and explicitly marks entries that still await final artifacts, including per-seed results.

REFERENCES

- Rajeev Alur, Rastislav Bodík, Garvit Juniwal, Milo M. K. Martin, Mukund Raghothaman, Sanjit A. Seshia, Rishabh Singh, Armando Solar-Lezama, Emina Torlak, and Abhishek Udupa. Syntax-guided synthesis. In *Proc. Formal Methods in Computer-Aided Design (FMCAD)*, pp. 1–8, 2013.
- Patrick Baudin, Pascal Cuoq, Jean-Christophe Filliâtre, Claude Marché, Benjamin Monate, Yannick Moy, and Virgile Prevosto. ACSL: ANSI/ISO C specification language. <https://frama-c.com/html/acsl.html>, 2021.
- Dirk Beyer. Competition on software verification (SV-COMP). <https://sv-comp.sosy-lab.org/>, 2023.
- Weining Cao, Guangyuan Wu, Tangzhi Xu, Yuan Yao, Hengfeng Wei, Taolue Chen, and Xiaoxing Ma. Clause2Inv: A generate-combine-check framework for loop invariant inference. *Proceedings of the ACM on Software Engineering*, 2(ISSTA):1009–1030, 2025. doi: 10.1145/3728920.
- Saikat Chakraborty, Shuvendu K. Lahiri, Sarah Fakhoury, Madanlal Musuvathi, Akash Lal, Aseem Rastogi, Aditya Senthilnathan, Rahul Sharma, and Nikhil Swamy. Ranking LLM-generated loop invariants for program verification. In *Findings of the Association for Computational Linguistics: EMNLP*, pp. 9164–9175, 2023. doi: 10.18653/v1/2023.findings-emnlp.614.
- Michael A. Colón, Sriram Sankaranarayanan, and Henny B. Sipma. Linear invariant generation using non-linear constraint solving. In *Proc. Int. Conf. on Computer Aided Verification (CAV)*, pp. 420–432. Springer, 2003.
- Patrick Cousot and Radhia Cousot. Abstract interpretation: A unified lattice model for static analysis of programs by construction or approximation of fixpoints. In *Proc. 4th ACM SIGACT-SIGPLAN Symp. on Principles of Programming Languages (POPL)*, pp. 238–252, 1977.

- Michael D. Ernst, Jeff H. Perkins, Philip J. Guo, Stephen McCamant, Carlos Pacheco, Matthew S. Tschantz, and Chen Xiao. The Daikon system for dynamic detection of likely invariants. *Science of Computer Programming*, 69:35–45, 2007.
- Emily First, Markus N. Rabe, Talia Ringer, and Yuriy Brun. Baldur: Whole-proof generation and repair with large language models. In *Proc. 31st ACM Joint European Software Engineering Conf. and Symp. on the Foundations of Software Engineering (ESEC/FSE)*, pp. 1229–1241, 2023.
- Cormac Flanagan and K. Rustan M. Leino. Houdini, an annotation assistant for ESC/Java. In *Proc. Int. Symp. of Formal Methods Europe (FME)*, pp. 500–517. Springer, 2001.
- Robert W. Floyd. Assigning meanings to programs. In *Proc. Symp. in Applied Mathematics: Mathematical Aspects of Computer Science*, volume 19, pp. 19–32, 1967.
- Pranav Garg, Christof Löding, P. Madhusudan, and Daniel Neider. ICE: A robust framework for learning invariants. In *Proc. Int. Conf. on Computer Aided Verification (CAV)*, pp. 69–87. Springer, 2014.
- C. A. R. Hoare. An axiomatic basis for computer programming. *Communications of the ACM*, 12(10):576–580, 1969.
- Zhechong Huang, Zhao Zhang, Zeyu Sun, Huifeng Sun, and Yingfei Xiong. Reinforcement learning with negative tests as completeness signal for formal specification synthesis. *arXiv preprint arXiv:2604.05820*, 2026.
- Adharsh Kamath, Aditya Senthilnathan, Saikat Chakraborty, Pantazis Deligiannis, Shuvendu K. Lahiri, Akash Lal, Aseem Rastogi, Subhajit Roy, and Rahul Sharma. Finding inductive loop invariants using large language models. *arXiv preprint arXiv:2311.07948*, 2023.
- Michael Karr. Affine relationships among variables of a program. *Acta Informatica*, 6(2):133–151, 1976.
- Zachary Kincaid, Jason Breck, Ashkan Forouhi Boroujeni, and Thomas Reps. Compositional recurrence analysis revisited. In *Proc. ACM SIGPLAN Conf. on Programming Language Design and Implementation (PLDI)*, pp. 248–262, 2017.
- Florent Kirchner, Nikolai Kosmatov, Virgile Prevosto, Julien Signoles, and Boris Yakobowski. Frama-c: A software analysis perspective. *Formal Aspects of Computing*, 27(3):573–609, 2015.
- Hung Le, Yue Wang, Akhilesh Deepak Gotmare, Silvio Savarese, and Steven C. H. Hoi. CodeRL: Mastering code generation through pretrained models and deep reinforcement learning. In *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Chang Liu, Xiwei Wu, Yuan Feng, Qinxiang Cao, and Junchi Yan. Towards general loop invariant generation: A benchmark of programs with memory manipulation. In *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2024a.
- Ruibang Liu, Minyu Chen, Ling-I Wu, Jingyu Ke, and Guoqiang Li. Enhancing automated loop invariant generation for complex programs with large language models. *arXiv preprint arXiv:2412.10483*, 2024b.
- Lezhi Ma, Shangqing Liu, Yi Li, Xiaofei Xie, and Lei Bu. SpecGen: Automated generation of formal program specifications via large language models. In *Proc. Int. Conf. on Software Engineering (ICSE)*, 2025.
- Kenneth L. McMillan. Interpolation and SAT-based model checking. In *Proc. Int. Conf. on Computer Aided Verification (CAV)*, pp. 1–13. Springer, 2003.
- Ido Pinto, Yizhak Yisrael Elboher, Haoze Wu, Nina Narodytska, and Guy Katz. Not all invariants are equal: Curating training data to accelerate program verification with SLMs. *arXiv preprint arXiv:2603.15510*, 2026.
- Gabriel Ryan, Justin Wong, Jianan Yao, Ronghui Gu, and Suman Jana. CLN2INV: Learning loop invariants with continuous logic networks. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2020.

- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Lloyd S. Shapley. A value for n -person games. In Harold W. Kuhn and Albert W. Tucker (eds.), *Contributions to the Theory of Games II*, volume 28 of *Annals of Mathematics Studies*, pp. 307–317. Princeton University Press, 1953.
- Xujie Si, Hanjun Dai, Mukund Raghothaman, Mayur Naik, and Le Song. Learning loop invariants for program verification. In *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, pp. 7751–7762, 2018.
- Xujie Si, Aaditya Naik, Hanjun Dai, Mayur Naik, and Le Song. Code2Inv: A deep learning framework for program verification. In *Proc. Int. Conf. on Computer Aided Verification (CAV)*, pp. 151–164. Springer, 2020.
- Anjiang Wei, Tarun Suresh, Tianran Sun, Haoze Wu, Ke Wang, and Alex Aiken. InvBench: Can LLMs accelerate program verification with invariant synthesis? *arXiv preprint arXiv:2509.21629*, 2025.
- Cheng Wen, Jialun Cao, Jie Su, Zhiwu Xu, Shengchao Qin, Mengda He, Haokun Li, Shing-Chi Cheung, and Cong Tian. Enchanting program specification synthesis by large language models using static analysis and program verification. In *Proc. Int. Conf. on Computer Aided Verification (CAV)*, pp. 302–328. Springer, 2024.
- Guangyuan Wu, Weining Cao, Yuan Yao, Hengfeng Wei, Taolue Chen, and Xiaoxing Ma. LLM meets bounded model checking: Neuro-symbolic loop invariant inference. In *Proc. IEEE/ACM Int. Conf. on Automated Software Engineering (ASE)*, 2024a.
- Haoze Wu, Clark Barrett, and Nina Narodytska. Lemur: Integrating large language models in automated program verification. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2024b.
- Chuanhao Yan, Fengdi Che, Xuhan Huang, Xu Xu, Xin Li, Yizhi Li, Xingwei Qu, Jingzhe Shi, Chenghua Lin, Yaodong Yang, Binhang Yuan, Hang Zhao, Yu Qiao, Bowen Zhou, and Jie Fu. Re:Form—reducing human priors in scalable formal software verification with RL in LLMs: A preliminary study on Dafny. *arXiv preprint arXiv:2507.16331*, 2025.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- Fanpeng Yang, Xing Li, Shuling Wang, Jie An, Zeyu Sun, Shenghua Feng, Wenhan Wang, Weiye Wang, Naijun Zhan, and Fanjiang Xu. How powerful are LLMs in generating formal program specifications? In *Proc. Int. Conf. on Machine Learning (ICML)*, 2026a.
- Fanpeng Yang, Xu Ma, Shuling Wang, Xiong Xu, Qinxiang Cao, Naijun Zhan, Xiaofeng Li, and Bin Gu. Integrating symbolic execution with LLMs for automated generation of program specifications. *arXiv preprint arXiv:2506.09550*, 2026b.
- Jianan Yao, Gabriel Ryan, Justin Wong, Suman Jana, and Ronghui Gu. Learning nonlinear loop invariants with gated continuous logic networks. In *Proc. ACM SIGPLAN Conf. on Programming Language Design and Implementation (PLDI)*, pp. 106–120, 2020.
- Shiwen Yu, Ting Wang, and Ji Wang. Loop invariant inference through SMT solving enhanced reinforcement learning. In *Proc. ACM SIGSOFT Int. Symp. on Software Testing and Analysis (ISSTA)*, pp. 175–187, 2023. doi: 10.1145/3597926.3598047.
- Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? In *Advances in Neural Information Processing Systems*, volume 38, 2025.

A SCOPE AND MOTIVATING EXAMPLE

We focus on numerical loops because their arithmetic equalities and inequalities, including polynomial relations, admit scalable automatic checking. This controlled setting factors out arrays, heaps, recursive data structures, auxiliary functions, and inductive predicates, allowing the study to isolate algebraic discovery and loop dynamics. Concretely, we consider one braced `while` loop over scalar C integers, including nonlinear updates. Even this restricted setting can hide nonlinear relations across coupled recurrences. For example:

```
1 void f(int k, int z) {
2   int x = 1, y = 1, c = 1;
3   while (c < k) { c++; x = x*z + 1; y = y*z; }
4   /*@ assert 1+x*z-x-z*y == 0; */
5 }
```

When visible, the assertion can be copied verbatim as an invariant; when hidden, its nonlinear relation must be recovered from the coupled updates.

B PROOFS OF COMPOSITIONALITY

This section contains all proofs for the formal composition claims in Section 4.1. Write $\mathcal{L} = (P, B, S)$, where P denotes loop-entry states, B the guard, and S the body transition.

B.1 CONJUNCTION OF VALID INVARIANTS

Proof of Proposition 1. For each j , validity gives $P \Rightarrow I_j$; hence $P \Rightarrow \bigwedge_j I_j = I$. Suppose $I \wedge B$ holds before one execution of S . Then $I_j \wedge B$ holds for every j , so preservation of I_j establishes every I_j afterward. Therefore I is preserved and is a valid invariant.

Because a conjunction implies each conjunct, $I \preceq I_j$. It satisfies $I \prec I_j$ precisely when the reverse implication fails. Since

$$I_j \Rightarrow I \iff I_j \Rightarrow \bigwedge_{\ell \neq j} I_\ell,$$

the stated condition follows. In particular, two valid invariants form a strict strengthening of both exactly when neither invariant implies the other; without this non-redundancy condition, conjunction is only guaranteed to be no weaker. $\square$

B.2 MUTUAL SUPPORT AMONG ARBITRARILY MANY CLAUSES

Proof of Proposition 2. The first premise is exactly establishment of I_C . For preservation, assume $I_C \wedge B$ before executing S . The second premise establishes each c_j afterward. Since every conjunct then holds in the same successor state, their conjunction I_C also holds. Thus $\{I_C \wedge B\}S\{I_C\}$, and I_C is valid.

No standalone preservation premise appears in this argument. A clause may use all other conjuncts in the pre-state, so clauses that fail preservation separately may still be preserved jointly. Establishment is different: $P \Rightarrow I_C$ implies $P \Rightarrow c_j$ for every j . Hence a candidate false at some loop-entry state cannot be rescued by conjunction. $\square$

The phenomenon is not limited to pairs. For any $m \geq 2$, consider the cyclic transition, where x'_i denotes the value of x_i after one execution of S ,

$$x'_1 = x_2, \quad x'_2 = x_3, \quad \dots, \quad x'_{m-1} = x_m, \quad x'_m = x_1$$

with entry condition $P \equiv \bigwedge_i x_i = 0$, and candidates $c_i \equiv x_i \geq 0$. Each c_i is established. It is not preserved alone: a state can satisfy $x_i \geq 0$ while its predecessor under the cyclic assignment is

negative. Nevertheless, the conjunction $\bigwedge_i x_i \geq 0$ is preserved because every successor coordinate is an old coordinate constrained by another conjunct. Thus all m candidates can fail standalone preservation while forming one valid, mutually supported invariant.

B.3 GREATEST JOINTLY INDUCTIVE CANDIDATE SUBSET

Let $C_0 = C$, and let $C_{t+1} \subseteq C_t$ be the set remaining after one ideal Houdini deletion round in Algorithm 1. Because C is finite, this descending sequence reaches the fixed point $H_{\mathcal{L}}(C)$.

Proof of Theorem 3. At the fixed point, every $c \in H_{\mathcal{L}}(C)$ satisfies establishment and preservation under the full conjunction $I_{H_{\mathcal{L}}(C)}$. Applying Proposition 2 shows that this conjunction is valid.

It remains to prove greatestness. Let $C' \subseteq C$ be any jointly inductive subset. We show by induction over deletion rounds that $C' \subseteq C_t$. The base case $C' \subseteq C_0 = C$ is immediate. Assume $C' \subseteq C_t$, and take any $c \in C'$. Joint establishment of C' gives $P \Rightarrow c$, so c cannot be deleted for establishment. Joint preservation gives

$$\{I_{C'} \wedge B\} S \{c\}.$$

Since $C' \subseteq C_t$, the antecedent $I_{C_t} \wedge B$ implies $I_{C'} \wedge B$. Strengthening a valid Hoare precondition preserves validity, so

$$\{I_{C_t} \wedge B\} S \{c\}$$

also holds. Therefore no member of C' is deleted in round t , proving $C' \subseteq C_{t+1}$. At termination, $C' \subseteq H_{\mathcal{L}}(C)$. Conjunction reverses set inclusion, hence $I_{H_{\mathcal{L}}(C)} \preceq I_{C'}$. $\square$

The theorem is deliberately candidate-relative: Houdini cannot synthesize a missing clause, so it need not recover the strongest invariant expressible in the full assertion language. It is also an ideal-oracle result. The inference-time Frama-C/WP backend may time out or fail to prove a valid obligation; such clauses are conservatively removed. The final retained conjunction remains accepted by the verifier, but completeness and greatestness are not claimed for these resource-bounded executions.

B.4 CLOSED FORM OF THE COVERAGE-GAME SHAPLEY VALUE

The allocation of Equation (1) is the Shapley value of the coverage game $v(S) = |\bigcup_{i \in S} R_i|/|\mathcal{N}|$ on the rollout group.

Proposition 4 (Closed form of the coverage game). *For the coverage game $v(S) = |\bigcup_{i \in S} R_i|/|\mathcal{N}|$, the Shapley value is $\phi_i = \frac{1}{|\mathcal{N}|} \sum_{n \in R_i} \frac{1}{f(n)}$, and $\sum_i \phi_i = v(\mathcal{G})$.*

Proof. Fix a trace n rejected by $f(n)$ rollouts. In a uniformly random ordering of the group, n contributes to the marginal gain of exactly one player—the first of those $f(n)$ rollouts to appear—and by symmetry each of them is first with probability $1/f(n)$. Player i therefore receives expected credit $1/f(n)$ for each $n \in R_i$ and zero otherwise. Summing over $n \in \mathcal{N}$ and normalizing by $|\mathcal{N}|$ gives the closed form; efficiency follows because each covered trace distributes exactly one unit of credit among its $f(n)$ rejectors. $\square$

C IMPLEMENTATION DETAILS

Table 5 lists the implemented defaults. Execution sampling is used only to compute reinforcement-learning rewards; it does not construct the synthetic SFT corpus. SFT synthesis, RL, and inference share the rollout-combine-filter implementation: the same annotation extraction, per-rollout response cap, normalization, candidate combination, iterative syntax filtering, conservative semantic deduplication, and Houdini pruning. SFT serializes the filtered combination, RL scores the responses that populate it, and inference restores Q only after filtering. Inference does not construct execution

Constant	Value	Role
runs requested	12	input executions; deterministic no-input programs collapse to one, and oracle-body sampling doubles the request
sampler seed	0	one canonical example set per reward request unless overridden
input tiers	20 fixed values	deterministic near-input schedule listed below; unconstrained tier candidates are clamped to $[-64, 64]$
far probes	3, plus 2 ordered multi-parameter probes	seed-hashed input-envelope coverage, magnitude ≤ 512
relation deltas	dense: $\pm\{1, 2\}$; terminal: $\pm\{1, 2, 3, 5, 8\}$	single-axis and pairwise perturbations; terminal bases require a witnessed final transition rather than a forward dense window
dense window	3	sampled successors required for a relation base
relation base cap	96	bases stratified across positives
range-perturbation steps	$\pm\{5, 8, 13, 21, 34\}$	candidates kept only outside the sampled range
post-exit steps	24	continuation length after a genuine exit (one negative trace per run)
negative-example budgets	320 (≤ 64 fallback) / 64 / 128	relational / post-exit / range examples; stable round-robin within each kind
iteration cap	2×10^6	divergence safety net (capped runs flagged)
(w_c, w_g)	(1.0, 0.3)	negative rejection / Shapley group-credit weights
w_d	0.02	cost per conservative duplicate; unique zero-coverage clauses are not penalized
K	20 clauses	response cap applied independently to every generated response
w_o	0.05	overflow cost per clause after K
Houdini iterations	≤ 500	greatest-inductive-subset pruning
WP configuration	Alt-Ergo, 30 s, Typed model	current implementation default and per-goal budget

Table 5: Defaults for execution sampling (§4.3), reward computation (§4.5), inference, and verification.

examples. The operators are shared, but RL applies Houdini once per response for credit assignment whereas SFT and inference apply it after union; Appendix K analyzes this ordering mismatch.

The exact near-input tier order is 0, 1, 2, -1, 3, 5, 8, -2, 10, 17, 4, -3, 25, 6, 40, 7, -5, 13, 50, 9.

C.1 NEGATIVE-TRACE SAMPLING

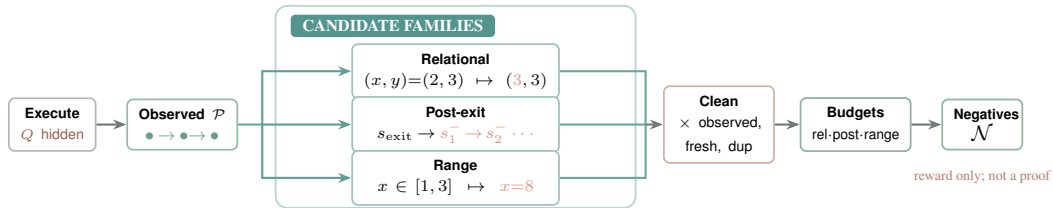


Figure 5: **Construction of target-hidden negative traces.** Concrete executions provide observed reachable states. Three transformations probe relational structure, behavior past a genuine exit, and sampled bounds. Conservative cleaning and per-family budgets produce $\mathcal{N}$, which is used only for RL reward computation and never exposes Q .

Algorithm 2 summarizes the sampler used for RL. Trace records real loop-head states and, after each genuine exit, continues the real loop body to obtain one post-exit trace. Perturb_Θ applies the configured single-variable and pairwise changes, while RangeEscape moves one variable just outside its sampled range. The exact perturbations and budgets are those in Table 5.

The three families are complementary: each rejects a different class of weak but valid invariants, and none subsumes another. Relational perturbations change one variable while fixing the rest, so they are rejected only by clauses that constrain relations among variables; pure bounds accept them. Post-exit continuations execute the real body past a witnessed exit and reach states that no finite perturbation of an observed state can produce, so only clauses that pin down the exit boundary reject them. Range escapes probe the boundary tightness of individual variables, which relational clauses may leave unconstrained. Coverage over their union therefore grades strength along three independent axes, and dropping any family leaves the corresponding weakness unpunished; Section 5.6 ablates these structured families against uniform random perturbations.

Algorithm 2 Target-hidden negative-trace sampling

Input: masked loop $\mathcal{L} = (P, B, S)$, input schedule U , configuration Θ
Output: reachable states $\mathcal{P}$, negative traces $\mathcal{N}$

- 1: $\mathcal{T}_{\text{exec}}, \mathcal{T}_{\text{post}}, c \leftarrow \text{Trace}(\mathcal{L}, U, \Theta) \triangleright c: \text{some run was capped}$
- 2: $\mathcal{P} \leftarrow \text{UniqueHeads}(\mathcal{T}_{\text{exec}})$; $M \leftarrow \text{SafeModifiedVars}(S)$
- 3: **if** $M = \emptyset$ or S has unsupported state **then return** $(\mathcal{P}, \emptyset)$
- 4: $\mathcal{P}_{\text{sample}} \leftarrow \text{StratifiedSample}(\mathcal{P})$
- 5: $C_{\text{rel}} \leftarrow \text{Perturb}_\Theta(\text{Dense}(\mathcal{P}_{\text{sample}}), M)$
- 6: **if** $\neg c$ **then** $C_{\text{rel}} \leftarrow C_{\text{rel}} \cup \text{Perturb}_\Theta(\text{Terminals}(\mathcal{T}_{\text{exec}}), M)$
- 7: $C_{\text{range}} \leftarrow \text{RangeEscape}(\mathcal{P}_{\text{sample}}, M, \mathcal{P}, \Theta)$ **if** $\neg c$, **else** $\emptyset$
- 8: $(C_{\text{rel}}, \mathcal{T}_{\text{post}}, C_{\text{range}}) \leftarrow \text{Clean}_{\mathcal{P}}(C_{\text{rel}}, \mathcal{T}_{\text{post}}, C_{\text{range}})$
- 9: $\hat{C}_{\text{rel}}, \mathcal{N}_{\text{post}}, \hat{C}_{\text{range}} \leftarrow \text{BudgetAndRoundRobin}(C_{\text{rel}}, \mathcal{T}_{\text{post}}, C_{\text{range}}, \Theta)$
- 10: $\mathcal{N} \leftarrow \{\langle s \rangle : s \in \hat{C}_{\text{rel}}\} \cup \mathcal{N}_{\text{post}} \cup \{\langle s \rangle : s \in \hat{C}_{\text{range}}\}$
- 11: **return** $(\mathcal{P}, \mathcal{N})$

$\text{Clean}_{\mathcal{P}}$ removes observed reachable states, possible fresh loop entries, duplicates, and unsupported candidates, using family priority relation, post-exit, then range. A cleaned post-exit continuation remains one trace; relation and range candidates become singleton traces. Within the relation budget, candidates that preserve the guard truth value and remain inside the sampled coordinate ranges are selected first.

C.2 GROUP-RELATIVE OPTIMIZATION DETAILS

For sampled sequences $o_{1:G} \sim \pi_{\theta_{\text{old}}}(\cdot \mid \mathcal{L})$, we normalize Equation 2 within its group,

$$\widehat{\text{Adv}}_i = \frac{r_i - \text{mean}_j r_j}{\text{std}_j r_j + 10^{-6}},$$

and use the clipped GRPO objective (Shao et al., 2024)

$$\mathcal{J}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_i \frac{1}{|o_i|} \sum_t \min \left(\rho_{i,t} \widehat{\text{Adv}}_i, \text{clip}(\rho_{i,t}, 1 - \varepsilon, 1 + \varepsilon) \widehat{\text{Adv}}_i \right) - \beta D_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) \right]. \quad (3)$$

Here $\rho_{i,t}$ is the token-level ratio to the rollout policy and π_{ref} is frozen. Duplicate detection normalizes comparison direction, equality sides, parentheses, immediate commutative operands, harmless identities, and Boolean association, but avoids transformations that require additional arithmetic assumptions. Unique zero-coverage clauses are not penalized because they may support another clause or the hidden target. Both coverage and Shapley credit lie in $[0, 1]$, so the unpenalized full reward lies in $[0, 1.3]$; duplicate and overflow costs can lower it. When the sampler retains no negatives, scoring falls back to binary inductiveness for a nonempty response. Combined-pool coverage is logged as a diagnostic but is not copied into every rollout reward.

D GENERATION PROMPT

The canonical evaluation template appears below; every target and non-contract comment is stripped before insertion, and general annotation rules are supplied by `system_prompt.txt`. We rebuild both training sets with this single prompt pair. From archival pools of 37,958 RL and 3,454 SFT records, the final artifacts retain 37,481 and 3,200 records, respectively. Fourteen records in each archive fall outside the scalar-integer, single-loop interface. A Frama-C kernel pass over 36,301 unique RL sources removes another 463 rows with C/ACSL parse or name-resolution errors. SFT cleaning removes 240 rows whose answers contain no surviving clause.

SFT answers are parsed into clauses, conservatively deduplicated, stripped of malformed, out-of-scope, tautological, and dominated constant-bound clauses, then checked by the same Frama-C/WP Houdini filter used by the method. The 5-second limit applies to each proof obligation. This reduces 32,883 archived clauses to 23,913 verified, non-tautological clauses, with at most 20 clauses per answer. A final conservative theorem-prover pass removes 431 universally true clauses from the verified set; five answers then become empty. For archived helper calls, 65 fixed-exponent occurrences are expanded into explicit multiplication; 14 resulting clauses remain in 14 final examples. Symbolic exponents are never approximated by finite expansion. Instead, when two equations share the same symbolic power, we eliminate that term to derive a power-free polynomial relation and verify it on the corresponding program; 70 irreducible, non-redundant relations remain in 68 examples. We reject reducible product equalities, which encode weak disjunctions and are often vacuous under initialization, and remove guarded copies when the same answer already contains their unconditional conclusion. The remaining 2,083 symbolic-power clauses cannot be represented equivalently under the released scalar interface and are removed. The artifact records every per-program rewrite and verdict. We also minimize loop-entry labels: an occurrence of `\at (v, LoopEntry)` is retained only when v is a local whose final pre-loop assignment is directly an `unknown*` call. Parameters are rewritten to `\at (v, Pre)`, deterministic local initializers are recursively expanded, and the rewritten clauses are rechecked by Houdini. This rewrites 4,214 unnecessary references and leaves 2,137 necessary references in 1,807 clauses across 348 answers; an independent audit finds no violation of the rule.

A final interface-alignment pass removes legacy integer casts, normalizes two Unicode comparison operators, and rechecks every clause against the inference-time scalar parser. Cast normalization exposes 68 duplicates and 98 algebraic tautologies; ten further clauses use expressions outside the released grammar. Two additional top-level formulas with an explicitly true implication or disjunction arm are removed structurally. For bounds-only answers, a conservative static analysis proposes seven two-variable conservation equalities only when both variables have a single unconditional constant update. Houdini/WP retains one and rejects six. The result remains 3,200 nonempty answers and has zero inference-interface violations.

Contract-level audits find no target clause or non-contract comment in any final RL or SFT model prompt, and all rows contain the same current system and user templates. The archival full source remains only in RL reward metadata; the service strips its target before sampling and inductiveness filtering. Likewise, model-visible evaluation sources contain no provenance or outcome comments. These cleaned artifacts require retraining; results from checkpoints trained on the archived prompts are not evidence for the final protocol.

Generation prompt

```
You are given a C function. Produce the strongest inductive ACSL loop
invariants that capture the loop's behavior (conservation/relational laws
+ tight progress bounds). Output ONLY invariant lines, each formatted
exactly as:
loop invariant <expr>;
Output at most 20 invariant lines.
```

```
Program:
{program}
```

The accompanying system prompt appears verbatim below (also released as `prompt/system_prompt.txt`).

System prompt

You are a formal verification expert specializing in ACSL loop invariant synthesis for Frama-C/WP.

LOOP INVARIANT DEFINITION

A loop invariant is a property that:

1. Holds before the first loop iteration (initialization).
2. Is preserved by every loop iteration when the loop guard is true (inductiveness).
3. Combined with loop exit ('!guard'), helps establish required loop-exit facts.

RULES

- Add one core conservation/relational equality that explains loop updates (state transition law).
- Add minimal progress bounds for loop counters and key arithmetic variables.
- Prefer strong invariants; avoid weak/tautological ones.
- Do not add unrelated invariants just to increase count.
- MUST NOT modify or rewrite any 'unknown()'/'unknownN()' call.
- '\at(v, Pre)' can ONLY be used with function parameters, never with local variables initialized inside the function.
- '\at(v, LoopEntry)' denotes the value of v at the start of the loop and can ONLY be used with local variables initialized before the loop (using v = unknown();).
- Loop guard is NOT an invariant; weaken strict guards to non-strict bounds at exit.
- Parenthesize mixed equality and relational expressions so their grouping is explicit.
- Use 'loop invariant' only as a line prefix, never inside expressions.
- Emit scalar invariant expressions only; do not declare predicates, logic functions, axioms, or lemmas.
- Do NOT use the ternary '? :' operator; split cases with '==>' instead.
- '^' is bitwise XOR in C/ACSL, not exponentiation --- write 'x*x*x' for x cubed.
- Do NOT place a C boolean directly in arithmetic ('x + (a > b)' is invalid); use '==>' to split cases.
- Only use variables declared in the given function.
- Operators: comparison '==', '!=', '<', '<=', '>', '>='; logical '&&', '||', '!', '==>', '<==>'; arithmetic '+', '-', '*', '/', '%', '<<', '>>'.

ACSL INVARIANT SYNTAX

Allowed expression forms (all verified by Frama-C/WP):

1. **Bounds**: '0 <= i', 'i <= n', 'x >= 1'
2. **Linear equality**: 'i + c == n', '2 * s == i * (i - 1)', 'x == q * y + r'
3. **Polynomial equality**: 'x == n * n * n', '6 * s == i * (i + 1) * (2 * i + 1)'
4. **Relational identity** (multi-var algebraic law): '1 + x * z - x - z * y == 0', 'p * s - r * q == 1'
5. **Implication**: 'i > 0 ==> s >= i', 'a != 0 ==> q >= 1'
6. **Modular**: 'i % 5 == 0', 'r >= 0 && r < y'
7. **Bitshift** (power-of-2 only): 'p == 1 << i', 's == (1 << i) - 1'
8. **Conjunction**: 'c >= 1 && c <= k'
9. **\at for params**: 'n == \at(n, Pre)' (function parameters only)

Forbidden:

- '^' for exponentiation (it is XOR) --- use repeated '*' or '<<' for powers of 2.
- '\product', '\sum', '\factorial' --- not ACSL scalar operators.
- '? :' ternary --- rewrite with '==>'.
- '\old(x)' --- use '\at(x, Pre)'.
- Type casts '(int)', '(long)' --- unnecessary in ACSL logic.
- Function calls, predicates, lemmas, axioms --- emit only scalar expressions.

Operators (complete list):

comparison: '==', '!=', '<', '<=', '>', '>='
logical: '&&', '||', '!', '==>', '<==>'
arithmetic: '+', '-', '*', '/', '%', '<<', '>>'

OUTPUT

Return ONLY invariant lines, one per line, formatted exactly as:

```
'loop invariant <property>;'
Do not return code fences, 'loop assigns', the surrounding C function,
or explanations.
```

E TRAINING INTERFACE

The training service is packaged as a self-contained, CPU-only container with gcc, the verifier, Why3, and its prover backend. It exposes POST /reward, accepting program source, an array of model responses, reward_variant in {binary, whole_coverage, base, base-shapley, full}, and sampler.negative_sampler in {random, structured}. The two switches are independent; the reward defaults to full and the sampler to structured. The response exposes the selected modes and Shapley credit explicitly:

```
{program, n_positives, n_negatives, filter_mode,
reward_variant, negative_sampler, reward_mode,
rollout_rewards[], base[], shapley_credit[],
redundant_clauses[], redundancy_penalty[],
generated[], accepted[], overflow[], overflow_penalty[],
rollouts[], batch_score}
```

Responses may be raw LLM text, invariant lists, or annotated code; parsing is uniform. An offline scorer provides the same generation reward over JSONL/Parquet rollout dumps.

F TRAINING–EVALUATION OVERLAP AUDIT

The 37,481 cleaned training records contain 35,824 unique source strings after whitespace normalization and include every program used for SFT. We compare this pool against all 832 evaluation files after removing Q , exactly as it is removed before model inference.

F.1 LAYERED MATCHING PROTOCOL

A single exact-string check can miss renamed or lightly rewritten copies. We therefore apply four increasingly aggressive program-level normalizations, all preserving `requires` clauses.

1. **Target-hidden source** removes Q and non-contract comments, then normalizes whitespace.
2. **Near-exact tokens** additionally anonymize the loop function name and tokenize the remaining source.
3. **Alpha-normalized source** renames identifiers in first-use order and maps the benchmark nondeterministic-function names to one token, while preserving constants, operators, statement order, and nesting.
4. **Constant-abstracted source** retains 0 and 1 but maps every other integer literal to one token. This catches copies changed only by variable names and nontrivial numeric constants.

We separately measure control-flow similarity. The *coarse profile* records whether the loop guard is constant, numbers of `if/else` nodes and compound conditions capped at three, and the presence of `break`, `continue`, `return`, `goto`, and nested loops. The *ordered control-flow skeleton* retains the order and brace nesting of control nodes and replaces each contiguous block of ordinary statements by one token. Its stricter variant also retains comparison and logical operators in branch conditions. These control-flow signatures intentionally discard data flow and arithmetic and are similarity diagnostics, not program-identity tests. The accompanying audit artifact records SHA-256 hashes of both training files and the complete evaluation corpus.

F.2 OVERLAP RESULTS

Representation	Shared signatures	Matched evaluation files
Target-hidden source	0	0 (0.0%)
Near-exact token skeleton	0	0 (0.0%)
Alpha-normalized source	0	0 (0.0%)
Constant-abstracted alpha source	0	0 (0.0%)
Coarse control-flow profile	21	762 (91.6%)
Ordered control-flow skeleton	21	524 (63.0%)
Skeleton + condition-operator shape	21	421 (50.6%)

Table 6: Layered training–evaluation overlap audit. Program-level representations preserve data flow; control-flow-only representations deliberately omit it.

No evaluation program matches the training pool at any program-level stage. In particular, overlap remains zero after all identifiers are alpha-renamed and after nontrivial constants are abstracted. This rules out exact copies, function-name or formatting variants, variable renamings, and copies changed only by numeric constants under the stated normalizations.

Control flow alone overlaps substantially. The coarse profiles of 762 of 832 evaluation files occur in training; preserving statement order and nesting reduces this to 524 files, and retaining condition-operator shapes reduces it to 421. This is expected for a scalar single-loop language: straight-line loops, one-branch loops, and `if--else` updates form a small reusable vocabulary. The contrast is the relevant observation: training covers common control-flow forms, but every match disappears once arithmetic and data-flow structure are retained. The observed control-flow overlap is therefore consistent with task coverage and does not by itself establish memorization.

Scope and residual risk. This audit covers the released fine-tuning data, not unknown base-model pretraining corpora. It may also miss algebraically equivalent updates, reordered independent statements, transformed loops, or distant semantic clones. We therefore report no overlap under the stated normalizations rather than claiming complete semantic independence.

G EXPERIMENTAL PROTOCOL DETAILS

The 37,481 RL records retain corpus-local identifiers but not per-record upstream-source identifiers. The separately retained SFT artifact contains 3,200 unique sources, of which 2,600 also occur in the RL artifact. We therefore characterize the training pool by content and audit its overlap, rather than inferring provenance from its IDs. The 832-file evaluation workload contains 316 linear programs (133 from CODE2INV, 84 from SyGuS 2019, and 99 from SV-COMP 2024), 50 nonlinear programs (30 from LIPuS and 20 from SV-COMP 2024), and the 466 integer programs in Loopy’s original 469-program corpus. Three floating-point programs are excluded. Unsupported inputs remain failures in the common denominator; because the collections reuse earlier benchmarks, the files are not claimed to be 832 independently sourced problems.

Before model invocation, we remove `assert` and `ensures` predicates, executable assertion calls, conventional error-label names, and non-contract comments, including source-provenance paths; preconditions and executable semantics remain visible. CRAFT evaluation runs use the canonical generation template, extractor, semantic deduplication, and Houdini implementation; external tools retain their own orchestration. Each model retains its serving interface and chat template. In CRAFT, API calls set `reasoning_effort=none`; all locally served checkpoints disable thinking. We set temperature 1.0 and top- p 1.0, without an additional top- k , repetition penalty, re-roll, or target feedback. The cap is 8,192 completion tokens for API models, including any provider-internal reasoning tokens, and 2,048 emitted tokens for local vLLM models. Each RQ1 curve reuses the same stochastic 100-response batch per task; no per-response sampling seed is fixed.

Training configuration. Table 7 separates settings recoverable from the released pipeline from fields that still require the original trainer manifest. The final submission must replace every TBD; reward and sampler constants are already reported in Table 5.

Field	Value	Scope or required provenance
Backbone ID and revision	TBD	exact Qwen3-8B repository and immutable revision
SFT optimizer / schedule	TBD	optimizer, learning rate, warmup, decay, and precision
SFT batch / duration	TBD	global batch, accumulation, epochs or updates, and sequence length
RL algorithm	GRPO	group-normalized objective in Equation 3
RL optimizer / schedule	TBD	optimizer, learning rate, warmup, decay, and precision
Rollout group size G	TBD	responses sharing one group-relative update
RL batch / duration	TBD	prompt batch, accumulation, effective batch, and update count
Clipping / KL	TBD	ϵ , β , and reference-policy update convention
Training seeds	TBD	seed for each reported checkpoint and ablation run
Hardware / software	TBD	GPU type and count, trainer, vLLM, CUDA, and framework versions
Local rollout decoding	non-thinking; $T = 1$, top- $p = 1$; 2,048 tokens	data synthesis, RL, and local evaluation

Table 7: Training configuration and required run-manifest fields.

H COMPLETE EXPERIMENTAL DATA

This section collects the per-model, per-stratum, and per-seed measurements behind the main aggregates. Entries marked TBD await the final evaluation artifacts; no missing value is used in the analysis.

H.1 PROBE CURVES

Table 8: Complete probe results. k_{95} is the smallest measured $k \in \{1, 10, 30, 50, 100\}$ at which $\text{combine@}k$ reaches 95% of its value at $k = 100$.

<i>Whole-response pass rates</i>						
Model	pass@1	pass@10	pass@30	pass@50	pass@100	
Qwen3-0.6B	0.58%	3.63%	6.88%	8.82%	11.54%	
Qwen3-4B	3.49%	8.05%	9.91%	10.69%	11.54%	
Qwen3-8B	3.55%	9.86%	12.78%	13.83%	15.02%	
Qwen3-14B	4.99%	10.37%	12.48%	13.40%	14.66%	
Qwen3-30B-A3B	5.00%	10.01%	11.52%	12.11%	12.98%	
Llama 3.1 8B	0.48%	3.38%	6.48%	8.19%	10.46%	
<i>Clause-combination rates</i>						
Model	combine@1	combine@10	combine@30	combine@50	combine@100	k_{95}
Qwen3-0.6B	9.50%	22.84%	27.40%	28.85%	29.21%	50
Qwen3-4B	23.08%	31.49%	34.13%	34.74%	36.06%	50
Qwen3-8B	26.32%	36.78%	38.22%	39.06%	39.78%	30
Qwen3-14B	30.17%	35.10%	37.14%	37.50%	37.74%	30
Qwen3-30B-A3B	28.73%	35.10%	35.82%	36.06%	36.54%	10
Llama 3.1 8B	18.03%	27.28%	28.25%	28.37%	28.37%	10

Table 9: Per-GPU decoding throughput of the base probe checkpoints, measured with vLLM on A800-80G GPUs ($n=100$ responses per task, temperature 1.0, `max_tokens=2048`).

Model	tok/s/GPU
Qwen3-0.6B	18,727
Qwen3-4B	7,445
Qwen3-8B	5,833
Qwen3-14B	3,168
Qwen3-30B-A3B	5,031
Llama 3.1 8B	5,135

H.2 SFT PROBE CURVES

The SFT checkpoints (0817, one epoch on the 3,200-example corpus) are evaluated under the same 832-task protocol as the base probe: 100 saved target-hidden responses per task at temperature 1.0, 5-second per-obligation prover budget, and the current prompt.

Table 10: Complete SFT probe results. k_{95} is the smallest measured $k \in \{1, 10, 30, 50, 100\}$ at which $\text{combine}@k$ reaches 95% of its value at $k = 100$.

<i>Whole-response pass rates</i>						
Model	pass@1	pass@10	pass@30	pass@50	pass@100	
Qwen3-4B + SFT	25.61%	53.66%	63.51%	67.47%	71.88%	
Qwen3-8B + SFT	25.98%	54.58%	63.84%	67.64%	72.12%	
Qwen3-14B + SFT	30.51%	57.90%	66.26%	69.45%	73.08%	
Llama 3.1 8B + SFT	17.76%	44.24%	53.29%	57.30%	62.26%	

<i>Clause-combination rates</i>						
Model	combine@1	combine@10	combine@30	combine@50	combine@100	k_{95}
Qwen3-4B + SFT	47.96%	70.07%	73.80%	73.92%	74.04%	30
Qwen3-8B + SFT	49.88%	74.04%	76.32%	76.80%	76.80%	10
Qwen3-14B + SFT	55.05%	73.68%	75.84%	76.44%	76.80%	10
Llama 3.1 8B + SFT	46.15%	64.66%	68.15%	68.63%	68.63%	30

Table 11: Per-GPU decoding throughput of the SFT checkpoints, measured with vLLM on A800-80G GPUs ($n=100$ responses per task, temperature 1.0, `max_tokens=2048`).

Model	tok/s/GPU
Qwen3-4B + SFT	7,445
Qwen3-8B + SFT	5,833
Qwen3-14B + SFT	3,168
Llama 3.1 8B + SFT	5,135

H.3 REINFORCEMENT-LEARNING RESULTS

Table 12: Complete comparison before and after RL. The Zero and SFT initializations are evaluated under the same decoding and verification configuration.

Path	Model	pass@1	combine@1	combine@10	combine@50	combine@100	k_{95}
Zero	Base 8B	3.55%	26.32%	36.78%	39.06%	39.78%	30
Zero	Base 8B + RL	TBD	TBD	TBD	TBD	TBD	TBD
SFT	SFT 8B	25.98%	49.88%	74.04%	76.80%	76.80%	10
SFT	SFT 8B + RL	TBD	TBD	TBD	TBD	TBD	TBD

H.4 ABLATION RESULTS

Reward variance is the mean within-group variance before GRPO normalization. Reachability collision is the fraction of sampled negative traces containing a state observed under an expanded execution sweep.

The reward variants correspond directly to the two levels of Equation 2. *Binary Inductiveness* assigns one iff the capped, normalized response is nonempty and every clause survives Houdini. *Whole-Rollout Strength* uses negative coverage only when every capped clause survives,

$$r_i^{\text{whole}} = \mathbf{1}[C_i^{\text{train}} = \text{set}(\text{dedup}(\bar{A}_i))] \frac{|R_i|}{|\mathcal{N}|} - w_o o(A_i).$$

Clause-Decomposed Strength removes this all-or-nothing indicator and scores the surviving C_i^{train} ; *Compositional Credit* additionally adds $w_g \phi_i$; and *Regularized Compositional Credit* additionally subtracts $w_d d_{\text{dup}}(A_i)$, yielding the full reward. These are `binary`, `whole_coverage`, `base`, `base_shapley`, and `full`, respectively.

Table 13: Reward-function ablation on Qwen3-8B (three-seed means, %); exact numbers for Figure 4. Best result per column among trained variants in bold.

Variant	pass@ <i>k</i>					combine@ <i>k</i>				
	1	10	30	50	100	1	10	30	50	100
Untrained 8B (reference)	3.55	9.86	12.78	13.83	15.02	26.32	36.78	38.22	39.06	39.78
Binary Inductiveness	8.65	10.84	12.09	12.66	13.46	1.80	5.89	8.53	9.98	12.02
Whole-Rollout Strength	10.80	12.25	12.81	13.10	13.46	22.24	24.88	25.72	25.96	26.56
Clause-Decomposed Strength	2.66	6.32	7.65	8.10	8.65	29.57	34.74	37.14	38.22	39.06
Compositional Credit	2.31	5.35	6.59	6.97	7.33	30.77	36.06	38.46	39.30	41.11
Regularized Compositional (ours)	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 14: Per-seed reward-ablation results on the 8B model. Seed columns and summary statistics report combine@10 verification rates.

Reward	Seed 1	Seed 2	Seed 3	Mean	Std.
Binary Inductiveness	TBD	TBD	TBD	TBD	TBD
Whole-Rollout Strength	TBD	TBD	TBD	TBD	TBD
Clause-Decomposed Strength	TBD	TBD	TBD	TBD	TBD
Compositional Credit	TBD	TBD	TBD	TBD	TBD
Regularized Compositional Credit	TBD	TBD	TBD	TBD	TBD

Table 15: Per-seed negative-trace sampler ablation on the 8B model. Seed columns and summary statistics report combine@10 verification rates.

Sampler	Seed 1	Seed 2	Seed 3	Mean	Std.
Random perturbation	TBD	TBD	TBD	TBD	TBD
Structured sampler	TBD	TBD	TBD	TBD	TBD

H.5 CROSS-MODEL MAIN RESULTS

Table 16: Complete main results by evaluation stratum (316 linear, 50 NLA, 466 Loopy, 832 overall programs), reproducing every pass@*k*/combine@*k* row from Table 1 and adding reasoning-on reference rows: *default* leaves reasoning_effort unset (the provider’s own default); *medium* sets it explicitly. GPT-5-mini has no setting that reaches true reasoning_effort=none, so it is omitted from Table 1 and appears here only under default reasoning.

Model	Reasoning	<i>k</i>	Linear	NLA	Loopy	All	Time / task	Tokens / task
GPT-5-nano	none	1	2.53/49.68	0.00/8.00	4.72/45.71	3.61/44.95	29.66 s	1,135.83
GPT-5-nano	none	10	17.72/67.72	4.00/10.00	19.74/71.24	18.03/66.23	190.35 s	4,322.38
GPT-5	none	1	24.05/62.03	2.00/10.00	24.46/63.52	22.96/59.74	18.22 s	1,230.92
GPT-5	none	10	53.16/71.20	10.00/8.00	57.73/68.88	53.13/66.11	75.53 s	12,250.89
Claude Sonnet-4.6	none	1	53.16/72.78	8.00/10.00	55.58/75.75	51.80/70.67	19.86 s	4,010.26
Claude Sonnet-4.6	none	10	43.67/66.77	10.00/12.00	51.07/64.81	45.79/62.38	66.94 s	13,071.64
DeepSeek-V4-Flash	none	1	16.14/62.66	6.00/12.00	21.67/61.59	18.63/59.01	35.90 s	1,209.49
DeepSeek-V4-Flash	none	10	47.47/72.15	8.00/8.00	55.15/72.53	49.40/68.51	105.24 s	13,580.48
DeepSeek-V4-Flash	default	1	57.91/62.03	4.00/8.00	52.15/54.51	51.44/54.57	166.98 s	11,084.44
GPT-5-mini	default	1	55.70/77.85	4.00/10.00	60.94/75.75	55.53/72.60	58.36 s	4,509.32
GPT-5	medium	1	77.85/80.38	14.00/14.00	80.47/80.47	75.48/76.44	82.88 s	5,289.00

All timing and token columns are full-workload means over 832 programs. Each $\text{pass}@k/\text{combine}@k$ pair at a given k is certified from the same saved k responses.

H.6 COMPARISON WITH SPECIALIZED TOOLS

Table 17: Archived target-hidden tool-comparison results reproducing the `reasoning_effort=none` rows from the main text alongside the `reasoning_effort=medium` archived rerun of the same methods; Daikon makes no model call, so it is reasoning-independent and listed once. Each cell reports the verification rate under the common file-level denominator.

Method	Reasoning	Linear / 316	NLA / 50	Loopy / 466	Overall / 832	Tokens / task	Time / task
AutoSpec	none	47.78%	6.00%	42.27%	42.19%	2,181.72	27.34 s
AutoSpec	medium	65.82%	8.00%	64.81%	61.78%	25,154.54	54.77 s
SESpec	none	28.80%	4.00%	33.05%	29.69%	22,081.61	68.50 s
SESpec	medium	56.96%	6.00%	49.57%	49.76%	44,807.05	117.31 s
Clause2Inv	none	20.89%	0.00%	0.00%	7.93%	1,056.25	8.41 s
Clause2Inv	medium	19.94%	2.00%	0.00%	7.69%	385.80	29.04 s
Daikon	—	23.10%	0.00%	21.67%	20.91%	0	4.89 s
LOOPY	none	20.25%	0.00%	19.74%	18.75%	14,513.37	147.16 s
LOOPY	medium	72.78%	12.00%	70.82%	68.03%	111,973.15	381.01 s
Naive	none	15.19%	2.00%	18.88%	16.47%	225.95	2.98 s
Naive	medium	48.73%	8.00%	47.42%	45.55%	4,493.37	28.87 s
CRAFT (pass@1)	none	2.53%	0.00%	4.72%	3.61%	1,135.83	7.86 s
CRAFT (pass@1)	medium	61.08%	14.00%	54.08%	54.33%	6,928.30	53.52 s
CRAFT (combine@1)	none	49.68%	8.00%	45.71%	44.95%	1,135.83	29.66 s
CRAFT (combine@1)	medium	64.24%	14.00%	63.09%	60.58%	6,928.30	66.60 s
CRAFT (combine@5)	none	64.87%	12.00%	63.73%	61.06%	2,161.52	91.78 s
CRAFT (combine@5)	medium	79.11%	14.00%	78.33%	74.76%	33,404.56	114.95 s
CRAFT (combine@10)	none	67.72%	10.00%	71.24%	66.23%	4,322.38	190.35 s
CRAFT (combine@10)	medium	80.70%	14.00%	80.69%	76.68%	68,953.35	143.67 s

The main text reports each method’s `reasoning_effort=none` row. The shared model-call parameters for the non-reasoning comparison are summarized separately in Table 18.

Table 18: Sampling parameters used in the non-reasoning target-hidden tool comparison.

Method	Sampling budget / task	Temperature / top- p	Completion cap	Reasoning
AutoSpec	8	1.0 / 1.0	8,192	none
SESpec	1 initial + up to 3 refinements per phase	1.0 / 1.0	8,192	none
Clause2Inv	adaptive; 1 per call	1.0 / 1.0	8,192	none
Daikon	no model call	—	—	—
Naive	1	1.0 / 1.0	8,192	none
CRAFT	1, 5, or 2×5	1.0 / 1.0	8,192	none
LOOPY	15 initial + up to 7 repairs	0.7 / 1.0	8,192	none

Token totals are means per task, with three accounting qualifications. AUTOSPEC, SESPEC, and CLAUSE2INV have token totals for 828, 826, and 821 tasks, respectively; their accuracy retains the 832-task denominator. The 466 unsupported CLAUSE2INV tasks make no model call and contribute zero tokens. The retained CRAFT pass@1 timing covers the 466 Loopy tasks. pass@1 and combine@1 reuse the same first model response.

I DETAILED LIMITATIONS

Program scope. Our implementation targets numerical programs with one scalar-integer loop. This scope follows the dominant setting in prior loop-invariant benchmarks, but excludes nested loops and invariants over arrays, heaps, or recursively defined predicates. Such invariants require richer representations and proof dependencies, while available training corpora contain too few examples to support the discriminative signal used here. Our state perturbations cannot yet construct reliably informative negatives for inductive predicates or interacting nested loops (Liu et al., 2024a).

Inference engine. In preliminary comparisons, locally evaluated checkpoints achieved lower verification rates under vLLM than under alternative backends despite nominally matched decoding settings. We nevertheless use vLLM for all local base and trained checkpoints because its throughput makes broad sampling practical and its consistent use avoids an additional train–test mismatch. API models retain their provider serving interfaces and chat templates under the explicitly reported decoding and reasoning controls. Local-model success rates may therefore be conservative rather than engine-independent ceilings.

Verifier and specification language. Our filtering and all final judgments use Frama-C/WP, and candidates use ACSL. Parser behavior, generated obligations, prover integration, and syntax affect which clauses survive. RCF is not intrinsically tied to this toolchain, but the quantitative conclusions may not transfer unchanged to other verifiers, specification languages, or source languages.

Finite candidate-pool approximation. Before Houdini, the combine implementation applies a lightweight AST-based filter to parse, normalize, and deduplicate candidate clauses. This filter is deliberately conservative and does not provide a completeness guarantee for the full ACSL expression language: a semantically useful clause that is not handled by its restricted parser can be omitted. In addition, `combine@n` caps the candidate union passed to verification at 300 clauses to keep WP cost bounded. When an extreme response pool exceeds this limit, later clauses may be truncated even if they add useful information. Both effects can make the reported `combine@n` underestimate an ideal unbounded implementation. They affect candidate recall, not the validity of reported successes, because every retained result must still pass the full Frama-C/WP final judge.

Feedback regime. We do not evaluate iterative optimization from verifier feedback. In pilot runs, models quickly reached valid but weak invariants: they passed inductiveness checks, but with Q hidden the verifier did not identify the missing behavioral strength. Refinement therefore yielded little marginal gain, which decreased further as larger initial rollout unions captured the easy corrections. Our results characterize target-independent training and finite-budget composition, not settings with rich counterexamples or target-directed repair.

J WHY THINKING IS DISABLED FOR LOCAL MODELS

For the non-reasoning API experiments reported in the main paper, GPT-family requests set `reasoning_effort=none`, compatible endpoints disable thinking explicitly, and Claude Sonnet disables adaptive thinking with a zero thinking budget. The archived reasoning-enabled tool results are labeled separately. For all locally served checkpoints, including the base probes and models trained in our pipeline, we disable chain-of-thought (CoT) during synthetic data construction, training rollouts, and evaluation. Applying the same setting at all three stages avoids a train–test distribution shift. This choice has two practical motivations.

CoT substantially increases generation cost. Our inference procedure samples many rollouts for each program. Explicit CoT adds reasoning tokens and generation latency to every rollout, so its cost is multiplied by the sampling budget. The required output, however, is only a short, machine-parseable set of ACSL clauses. We therefore omit explicit CoT to keep both data generation and multi-rollout inference efficient.

Invariant clauses compose, but CoTs do not. A single rollout may contain both valid and invalid candidate invariants. The invariant output is explicitly decomposable into clauses: Houdini can

1566 remove a failing clause while retaining the useful clauses, and the retained clauses can then be
1567 combined across rollouts. A CoT does not have this property. Correct and incorrect reasoning
1568 may be interleaved in one trace, with no reliable clause-level correspondence that would let us
1569 remove the reasoning associated with a rejected clause. Nor is there a principled way to merge the
1570 remaining fragments from different rollouts into one faithful CoT for the combined invariant set.
1571 Generating a new rationale after filtering would only produce a post-hoc explanation. Consequently,
1572 CoT supervision is not aligned with the clause-wise filter-and-combine target. For consistency, CoT
1573 remains disabled throughout the local-model pipeline.

1574
1575
1576
1577
1578
1579
1580
1581
1582
1583
1584
1585
1586
1587
1588
1589
1590
1591
1592
1593
1594
1595
1596
1597
1598
1599
1600
1601
1602
1603
1604
1605
1606
1607
1608
1609
1610
1611
1612
1613
1614
1615
1616
1617
1618
1619

K TRAINING–INFERENCE MISMATCH IN HOUDINI ORDERING

Let $\bar{A}_i$ be the normalized, capped clause sequence proposed by rollout i , and let $C_i^{\text{pf}} = \text{PF}_{\mathcal{P}}(\text{set}(\bar{A}_i))$. Recall that $\text{Filter}_{\mathcal{L}}(C)$ denotes the implemented filtering pipeline on candidate set C . SFT construction and inference first merge the candidates and then filter them:

$$C^{\text{infer}} = \text{Filter}_{\mathcal{L}}\left(\bigcup_{i=1}^G \text{set}(\bar{A}_i)\right). \quad (4)$$

This order allows a clause from one response to be preserved with the help of a companion clause proposed by another response.

RL requires a scalar reward for each rollout. The current implementation therefore runs Houdini independently and computes coverage, Shapley credit, and redundancy from

$$C_i^{\text{train}} = \text{Filter}_{\mathcal{L}}(C_i^{\text{pf}}). \quad (5)$$

It also evaluates $\text{Filter}_{\mathcal{L}}(\bigcup_i C_i^{\text{pf}})$ as a group-level diagnostic, but this shared result does not determine each rollout’s token-level advantage.

Why the two orders are not equivalent. Houdini is not distributive over union:

$$\text{Filter}_{\mathcal{L}}\left(\bigcup_i C_i^{\text{pf}}\right) \neq \bigcup_i \text{Filter}_{\mathcal{L}}(C_i^{\text{pf}}) \quad \text{in general.} \quad (6)$$

For example, a clause from one rollout may fail preservation in isolation but become inductive when conjoined with a supporting clause from another rollout. Thus, two clauses rejected by separate Houdini runs can belong to an inductive conjunction after union. Because the training reward filters each response first, it misses the positive credit associated with this cross-rollout support.

Why the practical effect is small. This is missed positive credit, not a dedicated penalty for non-inductiveness. The reward assigns coverage and Shapley credit to surviving clauses and penalizes duplicates and overflow. A unique clause that fails standalone Houdini incurs no additional penalty, although the absence of coverage can still give its rollout a lower group-relative advantage. At inference time, raw candidates from all responses are unioned before Houdini, so a complementary pair remains available and can be retained jointly. The mismatch therefore leaves inference-time cross-response support intact but may under-train clauses that depend on it. Pilot audits suggested that such rescues were rare, but we did not conduct a formal rescue-rate study; we therefore treat this as a qualitative observation, not an equivalence.

Why retain this approximation. Independent filtering gives a stable, parallelizable reward for every response. Assigning the same union-level score to all G responses is uninformative under group-relative normalization because the shared component cancels. Exact leave-one-out credit,

$$\Delta_i = V\left(\text{Filter}_{\mathcal{L}}\left(\bigcup_j C_j^{\text{pf}}\right)\right) - V\left(\text{Filter}_{\mathcal{L}}\left(\bigcup_{j \neq i} C_j^{\text{pf}}\right)\right),$$

requires at least $G + 1$ union-level Houdini evaluations per prompt and can still miss higher-order interactions. Shapley-style credit captures such interactions more faithfully but requires many subset evaluations. This would be the Shapley value of the union-and-Houdini verifier game, unlike the closed-form Shapley allocation of negative-set coverage used by our reward.

Implications and future correction. The ordering difference makes standalone Houdini a slightly imperfect reward surrogate for $\text{combine}@k$, but it does not alter inference-time union-and-Houdini. Future work can report the rescue rate represented by the gap between the two sides of Equation 6 and test leave-one-out or randomized-subset credit when that rate is non-negligible. For the present experiments, we retain the cheaper approximation and leave that cost–fidelity tradeoff to future evaluation.